



# Relational Graph enabled Incomplete Modality Learning via Decoupled Semantic Optimal Transport

Qinkai Yu<sup>a,b,c,†</sup>, Jing Li<sup>b,f,†</sup>, Feixiang Zhou<sup>d</sup>, Jianyang Xie<sup>d</sup>, Yitian Zhao<sup>e</sup>, Yalin Zheng<sup>d</sup>, Aline Villavicencio<sup>b</sup>, Gregory Y.H. Lip<sup>c</sup>, Yanda Meng<sup>a,\*</sup>

<sup>a</sup>Bioengineering Program, Biomedical Sciences Division (BioMed), King Abdullah University of Science and Technology (KAUST), Thuwal 23955, Saudi Arabia.

<sup>b</sup>Computer Science Department, University of Exeter, Exeter, UK.

<sup>c</sup>Liverpool Centre for Cardiovascular Science at University of Liverpool, Liverpool John Moores University and Liverpool Heart & Chest Hospital, Liverpool, UK.

<sup>d</sup>Eye and Vision Department, University of Liverpool, Liverpool, UK

<sup>e</sup>Ningbo Cixi Institute of Biomedical Engineering, Chinese Academy of Sciences, Cixi, China.

<sup>f</sup>Guangdong Provincial Key Laboratory of Multimodal Big Data Intelligent Analysis, South China University of Technology, Guangzhou, China.

## ARTICLE INFO

Article history:

**Keywords:** Incomplete Modality Learning, Optimal Transport, Semantic Decoupling, Relational Graph Neural Network

## ABSTRACT

Multimodal medical imaging provide complementary diagnostic information by integrating heterogeneous data sources. However, in real-world clinical practice, multimodal data is often incomplete due to equipment limitations, patient-specific constraints, or retrospective data collection difficulties, leading to the challenging problem of *incomplete-modality learning*. Existing multimodal approaches typically achieve strong performance under full-modality conditions but tend to overfit to dominant modalities during training. As a result, their representations become strongly modality-dependent and collapse when key modalities are absent. Moreover, the inherent heterogeneity across medical modalities makes cross-modal alignment and semantic consistency difficult when modalities are missing. To address these challenges, we propose an incomplete-modality learning framework that uses text-guided semantic decoupling to drive optimal-transport-based cross-modality alignment, thereby guiding graph-based completion of missing modality prototypes. Specifically, we propose a semantic–modality decoupling mechanism that explicitly separates diagnosis-related semantic representations from modality-specific appearance features via text guidance. Upon this, we propose an optimal transport alignment scheme that aligns class-consistent prototypes. We also construct modality-invariant semantic relations across heterogeneous modalities. Finally, we build a graph structure with constructed missing modality prototypes and semantic-invariant adjacency matrix, where we introduce a Gaussian sampling-based fusion mechanism to alleviate the problem of over-reliance on one dominant modality. Extensive experiments on three representative multimodal configurations demonstrate that the proposed framework achieves state-of-the-art performance and exhibits strong robustness and generalizability across diverse clinical modalities and diagnostic tasks.

© 2026 Elsevier B. V. All rights reserved.

## 1. Introduction

Multimodal medical imaging forms an essential foundation of modern clinical decision-making Li et al. (2024a). Different

\*Corresponding author: email: [Yanda.Meng@kaust.edu.sa](mailto:Yanda.Meng@kaust.edu.sa)

<sup>†</sup>Qinkai Yu and Jing Li contributed equally to this work.

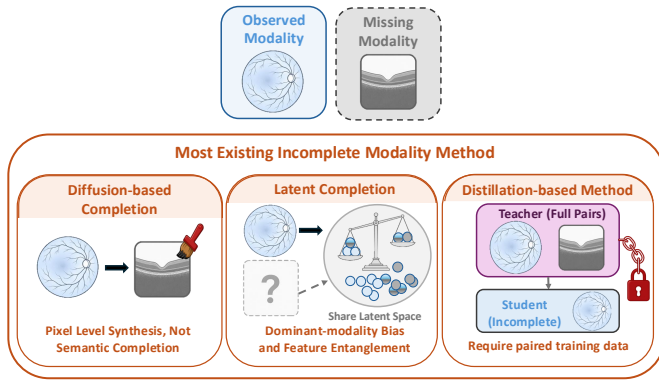


Fig. 1: Limitations of representative incomplete-modality learning paradigms. Given an observed modality and a missing modality, existing approaches commonly address incompleteness by synthesizing the missing modality appearance, completing features in a shared latent space, or distilling knowledge from fully paired modalities. However, diffusion-based completion mainly performs pixel-level appearance synthesis rather than semantic recovery; latent completion may suffer from dominant-modality bias and entangle diagnostic semantics with modality-specific features; and distillation-based methods often rely on paired training data and fixed modality configurations.

imaging and physiological acquisition modalities, such as CT, MRI, ultrasound, dermoscopy, and ECG prints, provide complementary views of the patient's condition, capturing diverse anatomical structures, tissue compositions, and functional dynamics Wu et al. (2023); Zhao et al. (2024). Such complementary information is routinely utilized by clinicians to localize lesions or evaluate disease severity Watanabe et al. (2022); Haribabu et al. (2023). Consequently, computational models that effectively integrate heterogeneous modalities hold the potential to enhance diagnostic accuracy, improve reliability under complex or ambiguous cases, and better reflect real-world clinical reasoning processes. However, in real-world clinical practice, multimodal acquisitions are often incomplete due to equipment availability, acquisition cost, patient-specific constraints, or retrospective data collection. This results in samples with partially missing modalities, a problem commonly referred to as *incomplete-modality learning* Wu et al. (2024). The inability to access all modalities simultaneously poses significant challenges for building robust diagnostic models that can generalize across heterogeneous and unbalanced modality distributions. Importantly, the same heterogeneity that makes multimodal medical data clinically complementary also makes incomplete-modality learning particularly difficult: when a modality is missing, the model may lose a distinct diagnostic view encoded under different acquisition physics, spatial scales, anatomical perspectives, and semantic characteristics, rather than merely losing redundant input information.

Addressing incomplete-modality learning remains challenging. Although many multimodal learning frameworks Wei et al. (2024); Hua et al. (2020); Zhao et al. (2024) achieve strong performance when all modalities are present during training, they often develop a strong dependency on one dominant modality. As a result, the learned representations tend to encode modality-specific appearance patterns rather than modality-invariant diagnostic semantics Du et al. (2023). When a modality be-

comes unavailable at inference, these models struggle to maintain cross-modal semantic consistency, leading to unstable feature representations and substantial degradation in downstream diagnostic performance. As reported in Table 1, this degradation can be severe under single-modality inference, where some multimodal or incomplete-modality baselines collapse to near-chance agreement, with Kappa values approaching zero or F1 scores dropping sharply in specific missing-modality settings. Such instability is not only caused by the absence of input information, but is further rooted in the intrinsic heterogeneity among medical modalities.

Given that a missing modality may correspond to a non-redundant diagnostic view, existing incomplete-modality methods generally attempt to approximate the unavailable evidence rather than simply discarding it. As illustrated in Figure 1, representative approaches can be broadly grouped into three paradigms: diffusion-based completion, latent space representation completion, and teacher-student distillation. Diffusion-based completion reconstruct missing modalities in pixel level Chen et al. (2024); Yang et al. (2024); Meng et al. (2024); Ye et al. (2025). Their supervision is largely defined at the pixel or image level, and therefore tends to emphasize texture, contrast, and structural fidelity. For medical diagnosis, such low-level reconstruction targets are not necessarily aligned with downstream clinical utility, since pixel-level similarity metrics may poorly reflect task-specific evidence for detection, segmentation, or classification Wohlrapp et al. (2026). More importantly, visual plausibility does not guarantee semantic consistency: disease-related cues such as subtle lesions, abnormal physiological patterns, or class-discriminative structures may be weakened or distorted during synthesis. Representation-level completion methods avoid explicit image synthesis by estimating missing-modality features in latent space. Representative methods include MoMKE Xu et al. (2024), ShaSpec Wang et al. (2023a), DrFuse Yao et al. (2024), and RIMA Yu et al. (2025). Compared with image-space generation, these methods are usually more efficient and can be integrated into end-to-end diagnostic models. Nevertheless, they often complete, align, or fuse heterogeneous modalities within a shared or partially shared representation space. In medical data, however, modalities may differ substantially in acquisition physics, spatial or temporal resolution, anatomical viewpoint, signal dimensionality, and diagnostic semantics. Compressing such heterogeneous information into a common latent space can cause the learned representation to be dominated by the modality with stronger or more stable discriminative cues. It may also entangle modality-specific acquisition patterns with disease-related semantics, so the completed representation can inherit the bias of the observed modality rather than faithfully recover the missing one. In addition, some representation-level frameworks are developed around particular modality configurations, such as paired 2D modalities in ShaSpec Wang et al. (2023a) or temporal signal-image fusion in DrFuse Yao et al. (2024), making them less flexible when applied to broader multimodal medical settings with different data forms. Another line of work relies on modality distillation, where an incomplete-modality student is trained to mimic a complete-modality or cross-modality

teacher Liu *et al.* (2025); Wang *et al.* (2023b). Although this paradigm can transfer useful disease-related knowledge, it typically requires paired multimodal training samples and a predefined teacher–student modality configuration. This requirement is restrictive in real clinical scenarios, where missingness may already exist in the training data and modality availability can vary across datasets, institutions, and tasks.

To address these challenges, we propose an incomplete-modality learning framework (shown in Figure 2) that uses text-guided semantic decoupling to drive optimal-transport-based cross-modality alignment, thereby guiding graph-based completion of missing modality prototypes. Our central motivation is that the semantic information shared across modalities should not be forced into a single modality-agnostic latent space. Although different modalities may describe the same disease category, their semantic feature distributions can still be heterogeneous because they are formed under different acquisition physics, spatial or temporal resolutions, anatomical viewpoints, and signal structures. Therefore, instead of learning one shared representation and several modality-specific residuals, we learn modality-conditioned semantic representations together with modality-specific variation representations for each modality. Specifically, we introduce a text-guided semantic-modality decoupling mechanism. For each modality, the encoder feature is decomposed into two branches: a semantic branch that captures diagnosis-related information under the corresponding modality, and a modality branch that captures acquisition- or appearance-specific variations. The text prompts are not used as an additional patient-specific modality. Instead, modality-conditioned class prompts provide class-level semantic anchors, encouraging the semantic branch to focus on diagnostic concepts while preventing it from drifting toward modality-specific texture, layout, or acquisition patterns. This design makes missing-modality completion more tractable, since the model only needs to recover the missing semantic component, rather than reconstructing unreliable modality-specific appearance or noise. We then use labeled optimal transport to build class-consistent correspondence between the decoupled semantic features of different modalities. Direct feature matching is unreliable for heterogeneous medical modalities, since their feature distributions may be very different. Labeled OT restricts the matching to samples from the same class and produces semantic prototypes from the learned transport plan. These prototypes serve as graph nodes, representing the diagnostic semantic states that should be recovered for the missing modality. In parallel, we construct graph edges using text-level OT over modality-conditioned class prompts. Since the prompt embeddings encode class semantics in a shared language space, the resulting adjacency matrix provides a modality-invariant relational prior that links the heterogeneous semantic distributions of different modalities. Based on these nodes and edges, we perform graph-based prototype completion. The reason for using a graph is that missing semantic features should not be predicted as isolated vectors. Their recovery should be consistent with both the class-level prototypes and the semantic relations between modalities. We therefore adopt a lightweight GCN to propagate information over the constructed graph and recover

the missing diagnosis-related semantic feature. Finally, we introduce a Gaussian sampling-based fusion strategy. The semantic feature is treated as the mean of the representation distribution, as it contains the main diagnosis-related evidence, while the modality-specific feature is treated as the standard deviation, reflecting acquisition-dependent variation or noise around the semantic content. For observed modalities, this sampling process helps model intra-class variation. For missing modalities, only the completed semantic feature is used for fusion, avoiding the use of uncertain modality-specific information. In addition, the proposed pipeline supports diverse modality configurations, including 1D–2D, 2D–2D, and 3D–2D clinical data.

The contributions of this work are as follows:

- We propose a text-guided semantic–modality decoupling mechanism that explicitly separates diagnosis-related semantic representations from modality-specific appearance features.
- We develop an optimal transport–based cross-modal alignment strategy, where labeled OT constructs class-consistent semantic prototypes and text-level OT derives modality-invariant semantic relations across heterogeneous modalities.
- We design a graph-based prototype completion module that treats OT-derived semantic prototypes as nodes and text-derived semantic relations as edges, enabling GCN-based recovery of missing diagnosis-related semantic features.
- We propose a Gaussian sampling–based modality fusion strategy that generates smooth stochastic representations and ensures stable inference.

Notably, this work is an extended version of our previous publication RIMA Yu *et al.* (2025), and we have made several substantial improvements and changes. First, compared with RIMA, which mainly performs missing-modality completion in a shared latent representation space, we reformulate incomplete-modality learning from a semantic-centric perspective. Specifically, we introduce a text-guided semantic–modality decoupling mechanism to explicitly separate diagnosis-related semantic representations from modality-specific appearance or acquisition variations, thereby reducing the risk that completed features inherit modality-dependent bias from the observed modality. Second, we replace direct latent completion with optimal-transport-based cross-modal semantic alignment. Labeled OT is used to construct class-consistent semantic prototypes from decoupled semantic features, while text-level OT derives class-level semantic relations from modality-conditioned prompts, providing a structured semantic prior for missing-modality recovery. Third, we introduce a graph-based prototype completion module, where OT-derived semantic prototypes are treated as graph nodes and text-derived semantic relations are used as graph edges, enabling relationally consistent recovery of missing diagnosis-related semantics. Finally, compared with RIMA, which was evaluated under a more limited modality configuration, we further validate the proposed framework across three representative multimodal settings, including 2D–2D, 3D–2D, and 1D–2D data,

to examine its robustness across different data forms, clinical modalities, and diagnostic tasks.

## 2. RELATED WORK

### 2.1. Multimodal Learning in Medical Imaging

Multimodal learning has emerged as a powerful paradigm in medical image analysis, as different imaging and physiological acquisition modalities provide complementary anatomical, functional, and semantic cues crucial for comprehensive clinical assessment Wu et al. (2023); Zhao et al. (2024). Typical examples include fundus photography and OCT for ophthalmic assessment, dermoscopic and clinical images for skin lesion analysis, and physiological signals together with visual recordings for cardiovascular diagnosis Watanabe et al. (2022); Wu et al. (2023); Liu et al. (2024). By integrating complementary observations from multiple sources, multimodal models can potentially improve diagnostic accuracy, robustness, and interpretability compared with unimodal systems Li et al. (2024a); Haribabu et al. (2023); Zhao et al. (2024).

Early multimodal fusion approaches Zuo et al. (2021); Hua et al. (2020); Tang et al. (2022) primarily relied on feature concatenation or attention-based aggregation, assuming that all modalities could be projected into a unified latent space. While effective in controlled settings, such designs often struggle to accommodate the substantial heterogeneity across modalities, including differences in imaging physics, spatial scale, appearance characteristics, and diagnostic semantics. As a result, naïve fusion may inadvertently mix modality-specific appearance patterns with diagnosis-related information, leading to suboptimal generalization when modality distributions shift.

To better capture modality diversity, a broad line of work has explored representation disentanglement strategies. Statistical normalization methods Xia et al. (2024); Wang et al. (2023c, 2024) aim to factor out modality style or dependent noise, whereas shared-private encoder architectures Wang et al. (2023a); Yao et al. (2024) explicitly separate common semantic information from modality-specific variations. More recent models, such as DMRNet Wei et al. (2024), employ probabilistic sampling to model modality-specific distributions. Approaches like STiL Du et al. (2025) similarly adopt shared-private designs, yet they enforce shared features to lie within a single latent space, which may obscure the intrinsic heterogeneity across modalities and weaken cross-modal consistency under distribution shifts.

Despite these advances, most existing multimodal learning methods are developed primarily for complete-modality settings, where all inputs are assumed to be available during both training and inference. This assumption is often violated in clinical practice, where multimodal acquisitions are frequently incomplete due to acquisition cost, protocol variation, equipment availability, or retrospective data collection. In such scenarios, the central challenge is no longer only how to fuse heterogeneous modalities, but also how to preserve diagnosis-relevant semantics when one or more modalities are absent.

Different from these approaches, our method performs semantic-guided disentanglement, which explicitly separates

diagnosis-related semantic representations from orthogonal modality-specific variations. Rather than forcing heterogeneous modalities into a single shared latent space, our framework aligns them through class-consistent optimal transport and reconstructs missing semantic features with graph-guided prototype completion, thereby improving robustness under incomplete-modality conditions.

### 2.2. Incomplete-Modality Learning in Medical Imaging

Incomplete-modality learning has gained increasing attention in medical imaging. Multimodal imaging data such as fundus photography, Optical Coherence Tomography (OCT), Magnetic Resonance Imaging (MRI), Positron Emission Tomography (PET), Computed Tomography (CT), and ultrasound are often complementary but not consistently available due to acquisition costs, protocol variations, or patient-specific limitations. The resulting modality incompleteness poses significant challenges for developing reliable diagnostic and analytic models. This motivates extensive research on robust missing-modality learning frameworks. Existing studies have explored this issue across various medical tasks, including segmentation Chen et al. (2024); Li et al. (2024b); Pipoli et al. (2025), image synthesis Meng et al. (2024), and disease grading Liu et al. (2025).

Current solutions to this problem can be broadly categorized into modality completion and modality distillation. Modality completion methods attempt to reconstruct missing information either at the image or representation level. Some modality completion approaches combined available inputs or synthesized the missing modality through image translation. For example, Yang et al. (2024) proposed Modal-mixup to complete the training datasets by randomly complementing same-category samples with missing modalities. Chen et al. (2024) addressed missing modalities in MRI segmentation by training U-Net models to generate the other two modalities from MRI. But they heavily relied on paired data and were prone to overfitting. More recently, diffusion-based modality synthesis methods have further improved the realism and flexibility of direct missing-modality reconstruction, especially for arbitrary missing scenarios Meng et al. (2024); Ye et al. (2025). However, these approaches still primarily optimize pixel-space fidelity and structural realism, which may overemphasize appearance reconstruction without guaranteeing preservation of diagnosis-related semantics under strong modality heterogeneity; moreover, their iterative denoising process introduces substantial computational overhead. To alleviate these limitations, recent studies have increasingly shifted from sample-level reconstruction to representation-level completion, which aims to infer latent features of missing modalities directly in feature space through multimodal fusion. Representative examples include MoMKE Xu et al. (2024), Shaspec Wang et al. (2023a), and RIMA Yu et al. (2025). Although more flexible and generally more efficient than image synthesis-based approaches, these methods are still constrained by the heterogeneity of medical modalities and often struggle to recover semantically faithful, modality-specific representations.

In contrast, modality distillation methods typically train a student model to mimic a teacher model that has access to the

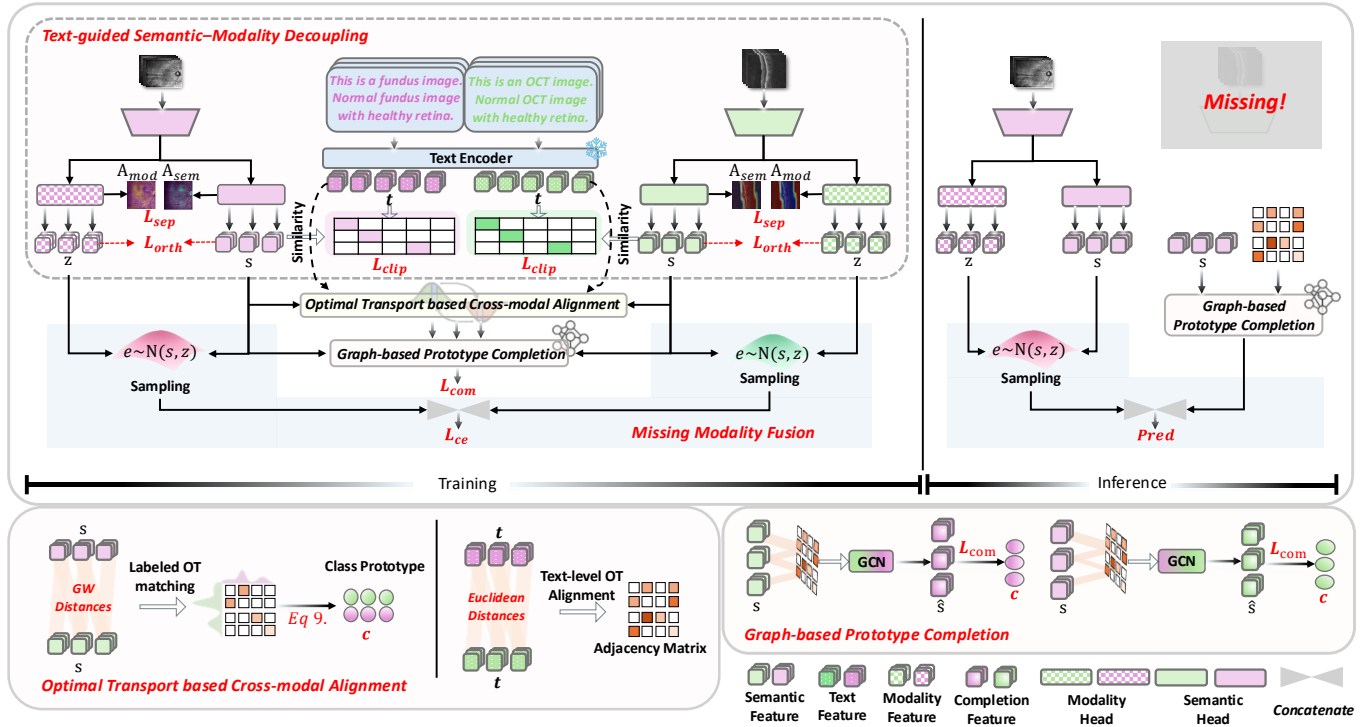


Fig. 2: Overview of the proposed framework. Our method consists of four key components: (1) **Text-guided Semantic-Modality Decoupling**, which separates diagnosis-relevant semantic features  $s$  and modality-specific appearance features  $z$ . This disentanglement is guided by text embeddings  $t$  using CLIP Radford et al. (2021)-based optimization. Furthermore, we employ a separation loss  $L_{sep}$  on the normalized “attention maps”  $A_{sem}$  and  $A_{mod}$ , and an orthogonality constraint  $L_{orth}$  between  $s$  and  $z$  to ensure strict independence between the semantic and modality subspaces; (2) **Optimal Transport-based Cross-modal Alignment**, where Labeled OT aligns the different modality’s semantic features  $s$  via Gromov-Wasserstein (GW) Peyré et al. (2019) distance to construct class prototypes  $c$ . Text-level OT calculates the transport plan between text embeddings  $t$  via Euclidean distance to build a semantic-invariant adjacency matrix  $W$  that describes the global semantic correspondence between modalities. (3) **Graph-based Prototype Completion**, which leverages a lightweight GCN to process semantic features  $s$  and the semantic-invariant adjacency matrix to reconstruct missing modality semantics. This reconstruction is supervised by the completion loss  $L_{com}$ , which enforces alignment between the completed features  $\hat{s}$  and the class prototypes  $c$ . (4) **Missing Modality Fusion**. During both training and inference, a Gaussian sampling-based fusion strategy is applied, where the semantic feature  $s$  is treated as the mean, and the modality feature  $z$  is treated as the standard deviation to generate stochastic representations. When a modality is missing, the completed semantic feature  $\hat{s}$  will be used for fusion.

complete multimodal data. The student model learns to approximate the missing-modality representations based on supervision from the teacher models. Several studies validated this paradigm in medical diagnosis tasks. For instance, Liu et al. (2025) introduced IMDR, leveraging a multimodal teacher model to guide a single-modality student model via disentangled representation learning. Wang et al. (2023b) distills disease-relevant knowledge from fundus images into an OCT-based student model. Although effective, these approaches typically assume fully paired multimodal training data, making them less practical for real-world clinical scenarios where incompleteness occurs in both training and inference.

Differently, our method can complete the decoupled semantic representations rather than reconstructing raw modality features. After separating semantic and modality-specific factors, we use labeled optimal transport to establish class-consistent cross-modal alignment, enabling reliable construction of semantic prototypes even under strong modality heterogeneity. Furthermore, a modality-invariant semantic adjacency matrix derived from text-level OT provides a stable relational prior, guiding a graph-based completion module to recover missing semantic features in a structurally consistent manner.

### 3. Methodology

Our framework proceeds in five stages. First, for each modality, we decouple the encoder feature into a diagnosis-related semantic component and a modality-specific component under text-guided supervision. Second, labeled Gromov-Wasserstein optimal transport establishes class-consistent cross-modal alignment and constructs soft semantic prototypes. Third, modality-conditioned class prompts are matched through text-level OT to produce a compact class-level semantic relation map, which serves as the adjacency matrix of the relational graph. Fourth, guided by this semantic prior, a graph completion module recovers missing semantic features from the available modalities. Finally, the completed or observed semantic features are fused with modality-specific features through Gaussian reparameterization for robust multimodal prediction.

#### 3.1. Background and Notation

Optimal Transport (OT) is used for cross-modal alignment Chen et al. (2022); Xu and Chen (2023); Gazdieva et al. (2023); Park et al. (2024); Luong et al. (2024). OT aims to find a transport map (or coupling)  $\mathcal{T}^*$ , signifying how much of a discrete

source distribution should be moved towards a discrete destination distribution to align the two distributions. In cross-modal alignment, the inherent uncertainty and heterogeneity across modalities make soft alignment suitable. It allows a sample to be partially transported to multiple targets and is typically solved using the Sinkhorn-Knapp algorithm Sinkhorn and Knopp (1967). The softness of the alignment is controlled by a regularization parameter  $\lambda_{\text{sinkhorn}}$ , where lower values result in harder alignment.

Consider  $\{x, y\} = \{x_n, y_n\}_n^N$  as a multimodal dataset that has  $N$  samples. Each sample  $x_n = \{x_n^1, x_n^2\}$  consists of inputs from 2 different modalities in this work, such as fundus images and OCT volumes; ECG images and ECG signals; clinical and dermoscopic color images. And the corresponding label is  $y_n \in \{1, 2, \dots, C\}$ , where  $C$  is the number of categories. For each modality  $m$ , a modality-specific encoder  $E^m(\cdot)$  extracts single-modality representations with an associated empirical distribution  $\mu^m$ , such as  $F^m = E^m(x^m) \in \mathbb{R}^{B \times C \times H \times W}$ , where  $B$  denotes the batch size,  $C$  is the number of feature channels,  $H$  and  $W$  represent the spatial height and width of the feature map. In addition, a text encoder  $T(\cdot)$  maps class prompts to semantic vectors. We denote the prompt set as  $\mathcal{P} = \{p_c\}_{c=1}^C$ , obtain the corresponding class-level semantic embeddings  $t_c = E_t(p_c) \in \mathbb{R}^{D_t}$ .

### 3.2. Text-guided Semantic–Modality Decoupling

Given a feature  $F^m$ , we produce two attention maps with lightweight convolutions of different kernel sizes to induce complementary receptive fields. Concretely, a semantic head is built with smaller kernels to capture fine-grained, lesion-centric structures. And a modality head is configured with larger kernels to identify global, modality-specific textures and styles. The outputs of the two heads are concatenated along the channel dimension and then passed through a channel-wise softmax, producing two normalized “attention maps” that form a mutually exclusive pair. This softmax normalization enforces complementarity: when the semantic attention becomes dominant at a spatial location, the modality attention is automatically suppressed, ensuring a clean and well-separated decomposition of semantic and modality-specific information. The resulting normalized maps are denoted as  $A_{\text{sem}}$  and  $A_{\text{mod}}$  (Shown in Figure 2). To ensure that the semantic and modality attentions focus on distinct and non-overlapping regions, we define a region separation loss  $\mathcal{L}_{\text{sep}}^m$ , which quantifies the spatial overlap between the two attention maps and enforces complementary, non-redundant focus across the two branches:

$$\mathcal{L}_{\text{sep}}^m = \frac{(A_{\text{sem}}^m \odot A_{\text{mod}}^m)_{\Sigma}}{(A_{\text{sem}}^m)_{\Sigma} + (A_{\text{mod}}^m)_{\Sigma} + \epsilon}, \quad (1)$$

where  $(\cdot)_{\Sigma}$  denotes summation over all spatial dimensions, and  $\epsilon$  is a small constant added for numerical stability. After that, the distinct  $A_{\text{sem}}$  and  $A_{\text{mod}}$  are then used to produce the final decoupled features:

$$s^m = \phi_{\text{sem}}(F^m \odot A_{\text{sem}}^m), \quad z^m = \phi_{\text{mod}}(F^m \odot A_{\text{mod}}^m), \quad (2)$$

where  $\odot$  denotes element-wise multiplication;  $\phi_{\text{sem}}$  and  $\phi_{\text{mod}}$  are 1x1 convolutional layers. After obtaining the spatially disen-

tangled features, we further regularize the projected representations to remain decorrelated in the embedding space. Specifically, a feature orthogonality loss ( $\mathcal{L}_{\text{orth}}^m$ ) is applied between the globally pooled semantic and modality features, where it encourages them to occupy near-orthogonal subspaces and prevents residual redundancy after projection. The  $\mathcal{L}_{\text{orth}}^m$  is defined as:

$$\mathcal{L}_{\text{orth}}^m = |\cos(s^m, z^m)|, \quad (3)$$

where  $s^m$  and  $z^m$  denote decoupled semantic feature and modality feature modality, respectively. The semantic branch, purified by spatial and feature-level disentanglement, is further anchored to diagnosis-relevant semantics through a CLIP Radford et al. (2021) based contrastive objective. Given a set of class prompts  $P^m = \{p_c^m\}_{c=1}^C$  and their text embeddings  $t_c^m = E_t(p_c^m)$ , the CLIP loss ( $\mathcal{L}_{\text{clip}}^m$ ) is defined as:

$$\mathcal{L}_{\text{clip}}^m = -\frac{1}{N} \sum \log \frac{\exp(\text{sim}(t_{y_n}, s_n^m)/\tau)}{\sum_{c=1}^C \exp(\text{sim}(t_c, s_n^m)/\tau)}, \quad (4)$$

where  $\text{sim}(\cdot, \cdot)$  denotes cosine similarity and  $\tau$  is the temperature parameter. Here  $t_{y_n}$  represents the textual embedding corresponding to the ground-truth class  $y_n$ . And  $s_n^m$  denotes the semantic feature of the  $n$ -th sample from modality  $m$ . This text-guided supervision provides an explicit semantic prior, ensuring that the disentangled semantic features remain aligned with clinically meaningful concepts rather than drifting toward arbitrary subspaces. The overall disentanglement objective ( $\mathcal{L}_{\text{decouple}}$ ) combines the spatial separation, feature orthogonality, and text-guided alignment losses:

$$\mathcal{L}_{\text{decouple}} = \sum_m (\mathcal{L}_{\text{sep}}^m + \mathcal{L}_{\text{orth}}^m + \mathcal{L}_{\text{clip}}^m). \quad (5)$$

These complementary objectives yield a structured and interpretable disentanglement that effectively separates diagnostic semantics from modality-specific variations, enabling robust multimodal learning under missing-modality conditions.

### 3.3. Optimal Transport–based Cross-modal Alignment

**Labeled Optimal Transport Matching.** To achieve cross-modal alignment under class supervision, we formulate Labeled Optimal Transport on top of the Gromov–Wasserstein (GW) Peyré et al. (2019) framework. Unlike the standard OT Cuturi (2013), where they align samples based on direct feature distances, Gromov–Wasserstein OT (GWOT) Peyré et al. (2019) compares relational geometry within each modality. This property makes it particularly suitable for heterogeneous medical data, where modalities (e.g., images, signals, volumetric scans) lie in distinct feature spaces. Instead of aligning absolute representations, GWOT Peyré et al. (2016) aligns the pairwise similarity structures, preserving relational consistency across modalities. Let  $\mu^m$  denotes the empirical distribution of modality Semantic feature  $s^m$ ,  $m \in \{1, 2\}$ . Formally, GWOT Peyré et al. (2016) seeks a transport plan  $T \in \mathbb{R}_+^{N \times N}$  by sinkhorn iterations Sinkhorn and Knopp (1967) that minimizes intra-domain distance discrepancies:

$$\min_{T \in \mathcal{T}_{\mu^1, \mu^2}} \sum_{i,j,k,l} |d_{ij}^1 - d_{kl}^2|^2 T_{ik} T_{jl} - \lambda_{\text{sinkhorn}} H(T) \quad (6)$$

Where  $d_{ij}^m$  denotes the pairwise distance between the  $i$ -th and  $j$ -th semantic features within modality  $m$ . And

$$\mathcal{T}_{\mu^1, \mu^2} = \{T \in \mathbb{R}_+^{N \times N} | T \mathbb{1}_N = \mu^1, T^\top \mathbb{1}_N = \mu^2\}, \quad (7)$$

is the set of feasible transport plans under marginal constraints;

$$\lambda_{\text{sinkhorn}} H(T) = \text{KL}(T \parallel \mu^1 \otimes \mu^2) = \sum_{i,k} T_{ik} (\log T_{ik} - 1), \quad (8)$$

is the entropy regularization with the scaling factor  $\lambda_{\text{sinkhorn}}$ . To ensure the alignment of corresponding semantic lesions across modalities, we constrain the transport to occur only between samples of the same class. The combination of the marginal constraints and this class-specific restriction yields a class-conditioned feasible set, defined as follows:

$$\tilde{\mathcal{T}}_{\mu^1, \mu^2} = \{T \in \mathcal{T}_{\mu^1, \mu^2} | T_{ij} > 0 \Rightarrow y_i^1 = y_j^2\}. \quad (9)$$

**Class Prototype Construction.** To obtain class-level representations under the Labeled OT alignment, we treat the transport plan  $T$  as a probabilistic mapping between samples of different modalities. Here element  $T_{i,j}$  denotes the probability of transferring  $s_i^1$  to  $s_j^2$ . And  $T_{i,\cdot}$  indicates the distribution over potential matches for the representation  $s_i^1$ . For a source semantic feature  $s_i^1$  in modality 1, its corresponding matching distribution (conditional probability) over modality 2 is given by  $p(j|i) = \frac{T_{i,j}}{\sum_{j'} T_{i,j'}}$ . The expected matched representation is computed as the soft prototype:

$$c_i^2 = \mathbb{E}[s_j^2 | s_i^1] = \sum_j p(j|i) s_j^2. \quad (10)$$

Similarly, prototypes  $c_j^1$  can be obtained in the reverse direction. These soft prototypes aggregate semantically consistent samples across modalities while preserving intra-class diversity.

**Text-level OT Alignment.** While the aforementioned labeled OT provides sample-level, class-consistent transport plans that facilitate prototype extraction, it does not explicitly encode the global semantic correspondence between modalities. This limitation arises because, despite text-guided decoupling, each modality's semantic features do not align well with their corresponding text prompts. The semantic spaces of different modalities still remain separated due to inherent heterogeneity. To establish a modality-invariant semantic bridge, we introduce a text-level OT alignment in the language space. Given text embeddings  $t^1 = \{t_c^1\}$  and  $t^2 = \{t_c^2\}$  that are derived from class prompts of the two modalities, we compute a feature-level coupling ( $T_v$ ) to align their semantic dimensions:

$$T_v = \arg \min_{T_v \in \mathcal{T}_{\mu_t^1, \mu_t^2}} \sum_{c=1}^C \sum_{c'=1}^C \mathbf{C}_{c,c'}^t (T_v)_{c,c'}, \quad \mathbf{C}_{c,c'}^t = \|t_c^1 - t_{c'}^2\|_2^2. \quad (11)$$

where  $\mathbf{C}^t \in \mathbb{R}^{C \times C}$  denotes the text-level semantic cost matrix between modality-conditioned class prompts, and  $T_v$  is the resulting text-level transport plan. Since the text space contains only  $C$  modality-conditioned class prompts for each modality, the transport problem is defined over a compact class-level prompt space. We solve this text-level OT problem without entropic regularization, yielding a relatively sharp coupling  $T_v$  between class prompts from different modalities. The resulting

transport plan captures cross-modality semantic relations at the class level and is used to construct the adjacency matrix  $W$  for graph-based prototype completion.

It is important to note that the text branch in our framework is not a patient-specific modality and does not rely on clinical reports or free-text annotations. Instead, it operates on modality-conditioned class prompts, where each prompt describes a disease category under a given modality. These prompts are only used to provide class-level semantic supervision for disentanglement and cross-modal relation modeling. Since the prompt set is predefined and fixed, the resulting semantic relation map can be computed from the prompt embeddings and reused at inference time without requiring any patient-specific textual input.

### 3.4. Graph-based Prototype completion and Missing Modality Fusion

**Graph-based Prototype Completion.** We perform cross-modal semantic completion guided by OT priors and the GCN translator. The text-level OT provides the semantic adjacency matrix  $W$  linking modality distributions, while the labeled OT ensures category-level consistency through class-prototype supervision. The adjacency matrix  $W$  derived from the text-level OT transport plan  $T_v$ , serves as a semantic prior to guide cross-modal prototype translation. The adjacency matrix  $W \in \mathbb{R}^{C \times C}$  encodes class-level semantic relations between modality-conditioned prompts, where  $I$  is the identity matrix and  $D$  is the degree matrix of  $W + I$ . Before being used by the GCN,  $W$  is symmetrically normalized as  $\tilde{W} = D^{-1/2}(W + I)D^{-1/2}$ . Given two source modality feature  $s_i^1$  and  $s_j^2$ , the GCN acts as a residual translator  $f_\theta$  that propagates semantic information through the graph and predicts corresponding class prototypes in the other modality:

$$\hat{s}_i^2 = f_\theta(s_i^1, \tilde{W}), \hat{s}_j^1 = f_\theta(s_j^2, \tilde{W}). \quad (12)$$

The completion features ( $\hat{s}_i^2$  and  $\hat{s}_j^1$ ) is supervised by their class prototype  $c_i^2$  and  $c_j^1$  obtained in the preview section. A bidirectional cosine-similarity loss enforces semantic alignment in the representation space rather than merely matching feature magnitudes:

$$\mathcal{L}_{com} = \sum_{i=1}^N [1 - \frac{\hat{s}_i^2 c_i^2}{\|\hat{s}_i^2\| \|c_i^2\|}] + \sum_{j=1}^N [1 - \frac{\hat{s}_j^1 c_j^1}{\|\hat{s}_j^1\| \|c_j^1\|}]. \quad (13)$$

**Missing Modality Fusion.** During training and inference, our goal is to maintain semantic completeness even when one modality is absent. That is because relying on single-modality prediction often suffers from biased decision boundaries due to strong inter-modality distributional heterogeneity (e.g., fundus vs. OCT, image vs. signal). For each modality  $m \in \{1, 2\}$ , we obtain the semantic feature  $s^m$  and modality feature  $z^m$  from the decoupled representations. If a modality is missing, we can use the completion semantic feature  $\hat{s}^m$  from graph-based prototype completion; otherwise, we sample a reparameterized feature  $e^m$  from the distribution, as shown below:

$$e^m = \begin{cases} \hat{s}^m, & \text{if modality } m \text{ is missing,} \\ s^m + \varepsilon z^m, \varepsilon \sim \mathcal{N}(0, I), & \text{otherwise,} \end{cases} \quad (14)$$

where  $\varepsilon$  is a random noise vector sampled from a standard normal distribution. This operation encourages smooth latent representations while allowing existing modalities to retain intra-class variability. The representations from both modalities (either existing modality or completed feature) are concatenated and passed through a linear classification head for final disease prediction. The overall training objective jointly optimizes three components:

$$\mathcal{L} = \mathcal{L}_{ce} + \beta \mathcal{L}_{decouple} + \gamma \mathcal{L}_{com}, \quad (15)$$

where  $\mathcal{L}_{ce}$  is a cross-entropy loss for disease prediction.

## 4. Experiment

### 4.1. Experimental Settings

#### 4.1.1. Datasets

To comprehensively evaluate the performance of the proposed models, we designed the following two experimental setups: 1) **Inter-Modality Missing**: Train on a complete training set and evaluate on a test set with a single modality. 2) **Complete-Modality Fusion**: Train on the complete training set and evaluate on the complete testing set; and 3) **Stochastic Missing-Ratio Sensitivity**: the missing ratio is varied from 30% to 70%, where one modality is randomly removed from each sample in both the training and test sets according to the specified ratio. We conducted extensive experiments on three large-scale multimodal datasets spanning diverse modality combinations to evaluate model robustness under missing-modality conditions. Specifically, the Harvard-30k AMD Luo et al. (2023) dataset involves 2D fundus images and 3D OCT volumes (2D–3D multimodality), the Derm7pt Kawahara et al. (2018) dataset includes paired dermoscopic and clinical photographs (2D–2D multimodality), and the MM-ECG dataset consists of 1D ECG signals and 2D ECG images (1D–2D multimodality). These datasets collectively cover a broad range of modality configurations and clinical scenarios.

**MM-ECG**: We reconstruct a new multimodal ECG dataset derived from the MIMIC-IV Johnson et al. (2023) subset of ECGInstruct Liu et al. (2024). It integrates 2D ECG images and 1D ECG time-series signals to facilitate multimodal analysis of cardiac rhythms. The dataset includes 8,795 samples categorized into three diagnostic classes: atrial fibrillation, sinus bradycardia, and sinus tachycardia. We open source the patient id of MM-ECG at <sup>2</sup>. The data are split into 7,035 training samples, 879 validation samples, and 880 test samples. For the 1D ECG modality, we preprocess it following Jin et al. Jin et al. (2025), resulting in a signal dimension of  $12 \times 1000$ . The 2D ECG images are generated using the ECG image synthesis tool proposed by Shivashankara et al. Shivashankara et al. (2024). We then resize the images to  $768 \times 768$  to preserve fine-grained waveform structures. Although the ECG images are derived from the same underlying signals and therefore do not introduce independent physiological information, they organize

the same cardiac activity into a different representational form. The raw 1D signals preserve lead-wise temporal waveform details, whereas the 2D ECG images explicitly present the sheet-level layout, cross-lead arrangement, and global morphological patterns of the recording. This setting is still meaningful for incomplete-modality learning, as ECG data may be available in clinical systems either as raw digital signals or as image-based reports/scans, while one representation may be unavailable due to acquisition, storage, or retrospective collection constraints. Therefore, MM-ECG is used in this work as a benchmark for learning robust diagnosis-related semantics across heterogeneous representations of the same physiological event under missing-modality conditions, rather than to claim strictly independent multimodal information.

**Derm7pt**: Derm7pt Kawahara et al. (2018) is used to evaluate computer-based image predictions of the 7-point checklist for skin lesion malignancy. We group the diagnosis labels in Derm7pt Kawahara et al. (2018) into five classes as originally proposed by Kawahara et al. Kawahara et al. (2018): BCC (basal cell carcinoma), MEL (melanoma), NEV (nevus), SK (seborrheic keratosis), and MISC (miscellaneous). The dataset contains two distinct types of 2D images: dermoscopic and clinical. In total, it includes 413 training samples, 204 validation samples, and 395 test samples. All images are resized to an input size of  $224 \times 224$ .

**Harvard-30k AMD**: Harvard-30k AMD is a curated subset of the Harvard-30k dataset Luo et al. (2023), which originally includes imaging data from 10,000 subjects diagnosed with age-related macular degeneration (AMD). We selected a high-quality subset consisting of 3,000 subjects followed by Liu et al. (2025), each with a scanning laser ophthalmoscopy (SLO) fundus image and a corresponding set of optical coherence tomography (OCT) scans. Each OCT volume has a size of  $200 \times 256 \times 256$  (where 200 denotes the number of OCT slices); Fundus images are resized to  $224 \times 224$ .

#### 4.1.2. Compared Methods

We compare our method with representative multimodal learning methods under both complete- and incomplete-modality settings. For a fair comparison, all applicable methods use the same modality-specific backbone encoders: 2D-ResNet-50 He et al. (2016) for fundus, dermoscopic, and ECG image analysis, R18-3D Tran et al. (2018) for volumetric OCT data, and a transformer encoder Vaswani et al. (2017) for 1D ECG sequences.

The compared methods include general multimodal fusion methods, i.e., B-EF Hua et al. (2020) and DMRNet Wei et al. (2024), as well as representative incomplete-modality learning methods, including MoMKE Xu et al. (2024), CLKD-IMRD Xu et al. (2023), IMDR Liu et al. (2025), DrFuse Yao et al. (2024), and RIMA Yu et al. (2025). Among them, DrFuse Yao et al. (2024) is specifically designed for 1D–2D multimodal data, and IMDR Liu et al. (2025) is tailored for 2D–3D multimodal scenarios. Other methods are modality-agnostic and can be adapted to different modality combinations by changing the backbone encoders.

<sup>2</sup><https://github.com/Qinkaiyu/MM-Ecg>.

Table 1: Comparison with other methods across three datasets, including Harvard-30k AMD Luo et al. (2023), MM-ECG, and Derm7pt Kawahara et al. (2018). The best results are **bolded**. Methods marked with † were not reimplemented; we cite their originally reported results for reference.

| Harvard-30k AMD Luo et al. (2023) |     |        |                   |                    |                          | MM-ECG                     |        |       |                    |                    |                   | Derm7pt Kawahara et al. (2018) |        |       |                    |                    |                   |
|-----------------------------------|-----|--------|-------------------|--------------------|--------------------------|----------------------------|--------|-------|--------------------|--------------------|-------------------|--------------------------------|--------|-------|--------------------|--------------------|-------------------|
| Model                             | OCT | Fundus | Kappa             | AUC                | F1                       | Model                      | Signal | Image | Kappa              | AUC                | F1                | Model                          | Clinic | Dermo | Kappa              | AUC                | F1                |
| Inter-Modality Missing            |     |        |                   |                    |                          |                            |        |       |                    |                    |                   |                                |        |       |                    |                    |                   |
| B-EF Hua et al. (2020)            | ✓   | ✓      | 37.58 ±0.66       | 79.12±3.15         | 61.44±7.92               | B-EF Hua et al. (2020)     | ✓      | ✓     | 87.29 ±0.88        | 97.42±1.49         | 91.35 ±0.62       | B-EF Hua et al. (2020)         | ✓      | ✓     | 23.72±6.43         | 70.80±4.76         | 50.76±7.58        |
| DMRNet Wei et al. (2024)          | ✓   | ✓      | 16.95±0.40        | 69.82±1.44         | 63.24±3.72               | DMRNet Wei et al. (2024)   | ✓      | ✓     | 54.80±1.18         | 90.23±1.95         | 70.57±0.40        | DMRNet Wei et al. (2024)       | ✓      | ✓     | 10.98±2.65         | 61.78±2.26         | 37.09±8.07        |
| MoMKE Xu et al. (2024)            | ✓   | ✓      | 15.92 ±4.53       | 70.49 ±3.07        | 61.24 ±2.81              | MoMKE Xu et al. (2024)     | ✓      | ✓     | 86.69 ±0.59        | 97.16±0.18         | 91.43 ±0.49       | MoMKE Xu et al. (2024)         | ✓      | ✓     | 50.21±13.51        | 87.88±2.46         | 68.91±7.69        |
| CLKD-IMRD Xu et al. (2023)        | ✓   | ✓      | 24.47 ±1.04       | 76.99 ±1.35        | 65.69 ±1.12              | CLKD-IMRD Xu et al. (2023) | ✓      | ✓     | 6.78±11.78         | 67.92±10.04        | 28.05 ±4.57       | CLKD-IMRD Xu et al. (2023)     | ✓      | ✓     | 49.12 ±12.46       | 83.71 ±4.69        | 70.11 ±6.57       |
| IMDR Liu et al. (2025)            | ✓   | ✓      | –                 | 80.48 <sup>†</sup> | <b>76.59<sup>†</sup></b> | DrFuse Yao et al. (2024)   | ✓      | ✓     | 85.94±2.13         | 96.71±0.15         | 91.07±1.24        | Shaspec Wang et al. (2023a)    | ✓      | ✓     | 14.43 ±7.79        | 65.25 ±7.93        | 38.90 ±5.59       |
| RIMA Yu et al. (2025)             | ✓   | ✓      | 44.91 ±8.19       | 83.75 ±2.52        | 74.46 ±2.88              | RIMA Yu et al. (2025)      | ✓      | ✓     | 87.43 ±0.61        | 97.27 ±0.26        | 90.68 ±0.27       | RIMA Yu et al. (2025)          | ✓      | ✓     | 30.88 ±7.36        | 82.05 ±1.79        | 54.70 ±7.09       |
| Our                               | ✓   | ✓      | <b>46.44±6.19</b> | <b>84.13±5.25</b>  | <b>74.49±2.07</b>        | Our                        | ✓      | ✓     | <b>87.85±0.53</b>  | <b>97.58±0.24</b>  | <b>91.78±0.36</b> | Our                            | ✓      | ✓     | <b>57.71± 2.01</b> | <b>87.90±1.30</b>  | <b>73.45±1.69</b> |
| B-EF Hua et al. (2020)            | ✓   | ✓      | 40.49±4.88        | 74.41±0.44         | 67.45 ±3.83              | B-EF Hua et al. (2020)     | ✓      | ✓     | 13.26±11.07        | 57.63± 8.71        | 12.28 ± 7.24      | B-EF Hua et al. (2020)         | ✓      | ✓     | 23.45 ±8.54        | 73.64 ±3.37        | 53.47±4.42        |
| DMRNet Wei et al. (2024)          | ✓   | ✓      | 8.76±0.50         | 62.09±1.20         | 55.33±0.43               | DMRNet Wei et al. (2024)   | ✓      | ✓     | 43.70±6.45         | 80.58±9.03         | 63.77±7.03        | DMRNet Wei et al. (2024)       | ✓      | ✓     | 6.70±2.17          | 56.30±1.21         | 43.45±9.57        |
| MoMKE Xu et al. (2024)            | ✓   | ✓      | 44.19 ±3.01       | 79.30 ± 0.24       | 70.22±3.64               | MoMKE Xu et al. (2024)     | ✓      | ✓     | 9.47 ±3.38         | 58.01±0.64         | 38.21 ±7.26       | MoMKE Xu et al. (2024)         | ✓      | ✓     | 16.23 ±14.13       | 69.84 ±5.18        | 49.33 ±8.47       |
| CLKD-IMRD Xu et al. (2023)        | ✓   | ✓      | 42.57±1.33        | 77.84 ±1.19        | 69.51 ±0.59              | CLKD-IMRD Xu et al. (2023) | ✓      | ✓     | 56.62±0.84         | 86.85±0.69         | 72.08 ±0.75       | CLKD-IMRD Xu et al. (2023)     | ✓      | ✓     | 14.59±4.15         | 69.15±0.38         | 45.55±6.82        |
| IMDR Liu et al. (2025)            | ✓   | ✓      | –                 | 72.69 <sup>†</sup> | <b>71.90<sup>†</sup></b> | DrFuse Yao et al. (2024)   | ✓      | ✓     | 71.09 ±12.23       | 91.17 ±1.69        | 81.62 ±3.38       | Shaspec Wang et al. (2023a)    | ✓      | ✓     | <b>43.18 ±2.32</b> | 80.82 ±0.77        | 46.48 ±1.88       |
| RIMA Yu et al. (2025)             | ✓   | ✓      | 37.75±5.19        | 79.21±6.81         | 70.59 ±2.52              | RIMA Yu et al. (2025)      | ✓      | ✓     | 4.20 ±5.94         | 58.30±3.58         | 21.01±15.77       | RIMA Yu et al. (2025)          | ✓      | ✓     | 18.56±7.70         | 76.90±0.83         | 49.86±5.20        |
| Our                               | ✓   | ✓      | <b>44.57±3.67</b> | <b>80.99±3.93</b>  | <b>70.39±2.98</b>        | Our                        | ✓      | ✓     | <b>83.56±4.96</b>  | <b>96.32±2.28</b>  | <b>89.49±5.14</b> | Our                            | ✓      | ✓     | 41.36±3.15         | <b>82.46±0.44</b>  | <b>64.63±1.87</b> |
| Complete-Modality Fusion          |     |        |                   |                    |                          |                            |        |       |                    |                    |                   |                                |        |       |                    |                    |                   |
| B-EF Hua et al. (2020)            | ✓   | ✓      | 39.67 ±5.19       | 79.12 ±0.42        | 69.25 ±3.85              | B-EF Hua et al. (2020)     | ✓      | ✓     | 86.32 ± 1.76       | 97.50 ±0.52        | 91.10 ± 0.39      | B-EF Hua et al. (2020)         | ✓      | ✓     | 44.84 ±1.46        | 82.49 ±0.37        | 66.49 ±0.81       |
| DMRNet Wei et al. (2024)          | ✓   | ✓      | 24.47 ±0.93       | 77.29 ±1.48        | 65.74 ±0.34              | DMRNet Wei et al. (2024)   | ✓      | ✓     | 81.65 ±4.15        | 96.03 ±1.21        | 88.16 ±2.35       | DMRNet Wei et al. (2024)       | ✓      | ✓     | 20.24 ±4.42        | 69.43 ±6.60        | 52.61 ±3.99       |
| MoMKE Xu et al. (2024)            | ✓   | ✓      | 43.18 ±5.07       | 80.82 ±0.65        | 71.48 ±2.78              | MoMKE Xu et al. (2024)     | ✓      | ✓     | 87.96 ±1.12        | 97.38 ±0.17        | 92.24 ±0.59       | MoMKE Xu et al. (2024)         | ✓      | ✓     | 53.62 ±3.31        | 89.12 ±1.06        | 71.60 ±2.77       |
| CLKD-IMRD Xu et al. (2023)        | ✓   | ✓      | 50.57 ±1.33       | 82.99 ±1.19        | 60.53 ±0.59              | CLKD-IMRD Xu et al. (2023) | ✓      | ✓     | 57.60 ±0.84        | 86.85 ±0.69        | 73.29 ±0.75       | CLKD-IMRD Xu et al. (2023)     | ✓      | ✓     | 56.23 ±4.46        | 86.06 ±3.69        | 73.14 ±4.57       |
| IMDR Liu et al. (2025)            | ✓   | ✓      | –                 | 83.10 <sup>†</sup> | 73.57 <sup>†</sup>       | DrFuse Yao et al. (2024)   | ✓      | ✓     | 85.94 ±1.69        | 96.72 ±0.18        | 91.07 ±1.40       | Shaspec Wang et al. (2023a)    | ✓      | ✓     | 32.44 ±2.46        | 75.61 ±5.90        | 59.29 ±1.54       |
| RIMA Yu et al. (2025)             | ✓   | ✓      | 49.45 ±1.26       | 83.20 ±0.33        | 74.33 ±0.54              | RIMA Yu et al. (2025)      | ✓      | ✓     | 87.43 ±0.26        | 97.34 ±0.39        | 92.14 ±0.60       | RIMA Yu et al. (2025)          | ✓      | ✓     | 48.06 ±2.44        | 86.08 ±1.08        | 68.09 ±1.67       |
| Our                               | ✓   | ✓      | <b>50.92±2.32</b> | <b>84.84±2.28</b>  | <b>75.28±0.79</b>        | Our                        | ✓      | ✓     | <b>88.10 ±0.95</b> | <b>97.58 ±0.30</b> | <b>92.74±0.50</b> | Our                            | ✓      | ✓     | <b>58.59 ±0.68</b> | <b>90.35 ±1.12</b> | <b>74.83±0.49</b> |

#### 4.1.3. Implementation Details

All OT based computations are implemented using the *perturb-OT* library Ryu et al. (2024). The model is trained for 150 epochs with a batch size of 32 on four NVIDIA RTX4090 GPUs. We adopt the AdamW Loshchilov and Hutter (2017) optimizer with an initial learning rate of  $1 \times 10^{-4}$ . The loss weighting coefficients in Equation 15 were selected via grid search, and the optimal values were fixed to  $\beta = 2.75$  and  $\gamma = 5$  for all experiments. For the main results, each experiment was repeated three times with different random seeds, and we report the mean and standard deviation.

#### 4.1.4. Evaluation Metrics

We evaluate the model performance using three commonly adopted metrics: Cohen’s Kappa (Kappa), Weighted F1-Score (F1) and Area Under the Curve (AUC). Notably, F1 can balance precision and recall.

#### 4.2. Main Result

Table 1 reports a comprehensive comparison of our proposed model against recent state-of-the-art multimodal methods or incomplete-modality learning methods across three multimodal datasets. Overall, the proposed method shows clear and consistent advantages in both missing-modality and complete-modality settings. Our model consistently demonstrates superior performance, outperforming all competitors on key metrics such as Kappa, F1 and AUC.

**Inter-Modality Missing.** Existing multimodal fusion methods often show strong performance only when all modalities are available, but degrade sharply once a modality is missing. This behaviour is evident in methods such as DMRNet Wei et al. (2024), MoMKE Xu et al. (2024), and CLKD-IMRD Xu et al. (2023), which perform reasonably well under complete fusion but drop dramatically in missing-modality settings (*e.g.*,

*MoMKE Xu et al. (2024) falls by 20–40% in Kappa and F1 when fundus-only on Harvard-30k AMD Luo et al. (2023)).* In contrast, our framework maintains stable and competitive performance across all missing-modality scenarios. For instance, on the Harvard-30k AMD Luo et al. (2023) dataset, our method improves the average AUC by 1.35%, and the Kappa by 9% compared to RIMA Yu et al. (2025) in both missing settings. A similar trend is observed on MM-ECG and Derm7pt Kawahara et al. (2018), where our approach demonstrates consistently higher performance in both inter-modality missing scenarios. When only one modality is available, the proposed prototype completion provides a semantically aligned substitute feature while avoiding the interference of modality features in classification decisions. The reparameterized sampling further prevents the model from collapsing into single-modality dominance, ensuring that each modality learns useful diagnostic semantics independently. Furthermore, our method achieves a significant performance improvement compared to the baseline unimodal method. On Harvard-30k AMD Luo et al. (2023), the proposed framework improves the OCT AUC score by 13% and yields a 10% increase for the fundus modality. For Derm7pt Kawahara et al. (2018), the clinical and dermoscopy modalities exhibit F1 improvements of 15% and 10%, respectively. This demonstrates that our model can effectively utilize complementary multimodal information to enhance unimodal inference performance.

**Complete-Modality Fusion.** In the complete-modality setting, our method also achieves the best overall performance across all three datasets, as shown in Table 1. For example, on Harvard-30k AMD Luo et al. (2023), our approach surpasses RIMA Yu et al. (2025) by 1.6% AUC and 1.0% F1. On Derm7pt Kawahara et al. (2018), we improve Kappa by 3.1% over MoMKE Xu et al. (2024) and F1 by 2% over CLKD-IMRD Xu et al. (2023). Similar advantages are consistently ob-

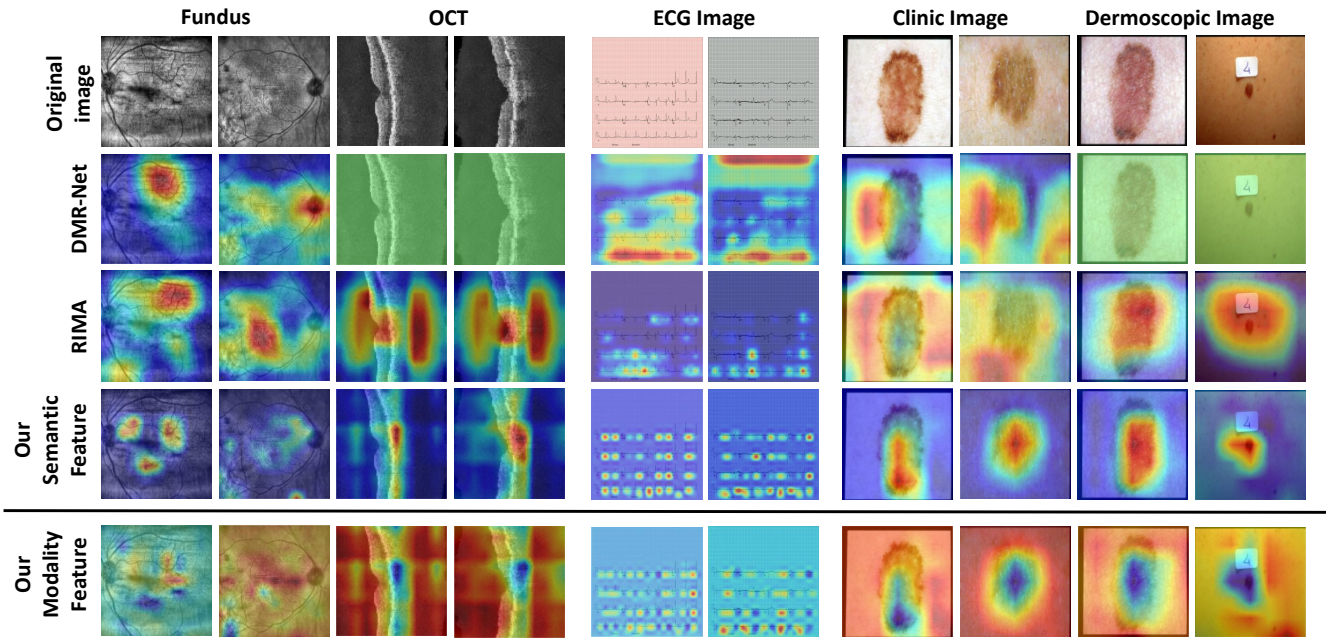


Fig. 3: **Visualization of text-guided semantic-modality decoupling.** We compare our method with two representative method: DMRNet Wei et al. (2024), a typical multi-modal learning framework, and RIMA Yu et al. (2025), a representative approach for incomplete-modality learning across Harvard-30k AMD Luo et al. (2023), MM-ECG and Derm7pt Kawahara et al. (2018) datasets. The semantic features  $s^m$ , and modality features  $z^m$  are visualized, including Fundus, OCT, ECG, Clinical, and Dermoscopic images. For OCT, which is a 3D volumetric modality, we visualize the representative B-scan slice. Note that ECG signals are not visualized owing to the architectural design of the model.

Table 2: Ablation study across three datasets, including Harvard-30k AMD Luo et al. (2023), MM-ECG, and Derm7pt Kawahara et al. (2018). The best results are **bolded**.

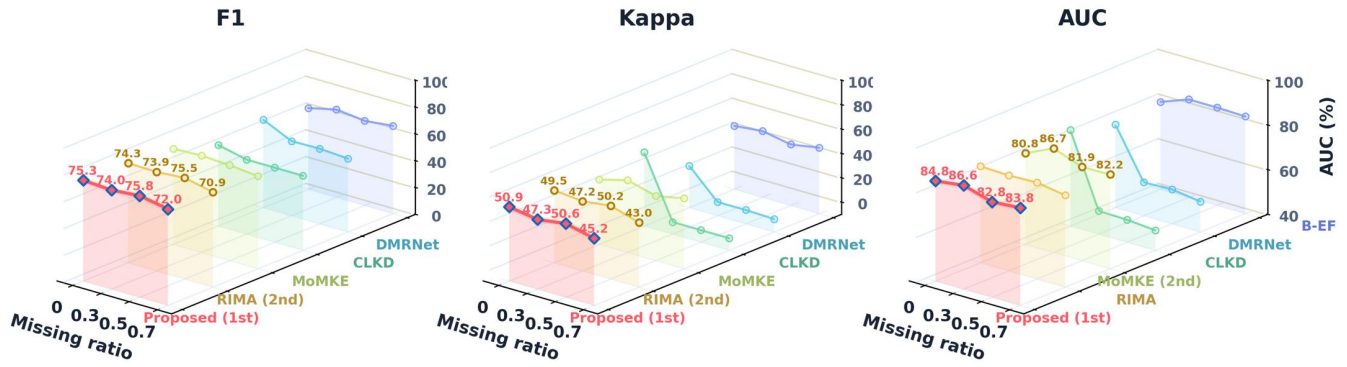
|                          | Harvard-30k AMD Luo et al. (2023) |        |              |              |              | MM-ECG |       |              |              |              | Derm7pt Kawahara et al. (2018) |        |              |              |              |
|--------------------------|-----------------------------------|--------|--------------|--------------|--------------|--------|-------|--------------|--------------|--------------|--------------------------------|--------|--------------|--------------|--------------|
| Model                    | OCT                               | Fundus | Kappa        | AUC          | F1           | Signal | Image | Kappa        | AUC          | F1           | Clinic                         | Dermo. | Kappa        | AUC          | F1           |
| Inter-Modality Missing   |                                   |        |              |              |              |        |       |              |              |              |                                |        |              |              |              |
| w/o Decoupling           |                                   | ✓      | 13.52        | 69.58        | 60.26        |        | ✓     | 87.11        | 95.92        | 91.35        |                                | ✓      | 21.85        | 69.53        | 53.11        |
| w/o OT                   |                                   | ✓      | 44.98        | 83.61        | 73.29        |        | ✓     | 87.46        | 96.92        | 92.61        |                                | ✓      | 46.72        | 84.38        | 70.02        |
| w/o GCN                  |                                   | ✓      | 45.07        | 83.81        | 74.22        |        | ✓     | 88.76        | 97.38        | 92.81        |                                | ✓      | 44.97        | 80.01        | 69.86        |
| Ours                     |                                   | ✓      | <b>46.44</b> | <b>84.13</b> | <b>74.49</b> |        | ✓     | <b>88.85</b> | <b>97.58</b> | <b>93.01</b> |                                | ✓      | <b>57.71</b> | <b>87.90</b> | <b>73.45</b> |
| w/o Decoupling           | ✓                                 |        | 20.01        | 68.27        | 55.98        | ✓      |       | 72.55        | 91.36        | 85.93        | ✓                              |        | 4.53         | 59.02        | 43.55        |
| w/o OT                   | ✓                                 |        | 35.62        | 79.23        | 68.76        | ✓      |       | 83.04        | 96.12        | 87.69        | ✓                              |        | 34.90        | 79.82        | 61.05        |
| w/o GCN                  | ✓                                 |        | 34.09        | 78.46        | 68.67        | ✓      |       | 83.23        | 95.90        | 89.28        | ✓                              |        | 37.35        | 80.88        | 61.94        |
| Ours                     | ✓                                 |        | <b>44.57</b> | <b>80.99</b> | <b>70.39</b> | ✓      |       | <b>83.56</b> | <b>96.32</b> | <b>89.49</b> | ✓                              |        | <b>41.36</b> | <b>82.46</b> | <b>64.63</b> |
| Complete-Modality Fusion |                                   |        |              |              |              |        |       |              |              |              |                                |        |              |              |              |
| w/o Decoupling           | ✓                                 | ✓      | 38.96        | 75.34        | 54.22        | ✓      | ✓     | 86.39        | 95.02        | 89.99        | ✓                              | ✓      | 51.32        | 86.12        | 69.81        |
| w/o OT                   | ✓                                 | ✓      | 46.62        | 84.14        | 74.23        | ✓      | ✓     | 86.61        | 96.28        | 91.43        | ✓                              | ✓      | 53.89        | 85.95        | 70.60        |
| w/o GCN                  | ✓                                 | ✓      | 46.66        | 84.10        | 73.98        | ✓      | ✓     | 87.75        | 97.36        | 91.80        | ✓                              | ✓      | 53.09        | 86.01        | 72.92        |
| Ours                     | ✓                                 | ✓      | <b>50.92</b> | <b>84.84</b> | <b>75.28</b> | ✓      | ✓     | <b>88.10</b> | <b>97.58</b> | <b>92.74</b> | ✓                              | ✓      | <b>58.59</b> | <b>90.35</b> | <b>74.83</b> |

served in MM-ECG, where our method outperforms RIMA by 0.3% in AUC and 0.6% in F1, and exceeds MoMKE by 0.25% and 0.5% on the same metrics. These results validate that our model does not merely align modalities for missing-modality inference; it also establishes a more reliable multimodal fusion

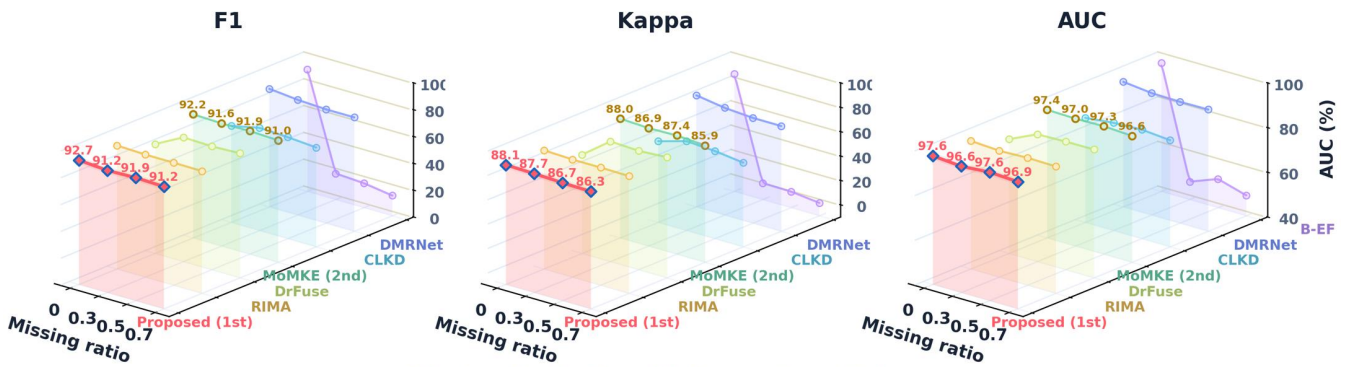
representation that enhances diagnostic performance when all modalities are available.

**Stochastic Missing-Ratio Sensitivity.** We further studied the sensitivity of different methods to different modality missing ratios, ranging from 30% to 70%. In this setting, one

## Missing-Ratio Sensitivity on Harvard-30k AMD



## Missing-Ratio Sensitivity on MM-ECG



## Missing-Ratio Sensitivity on Derm7pt

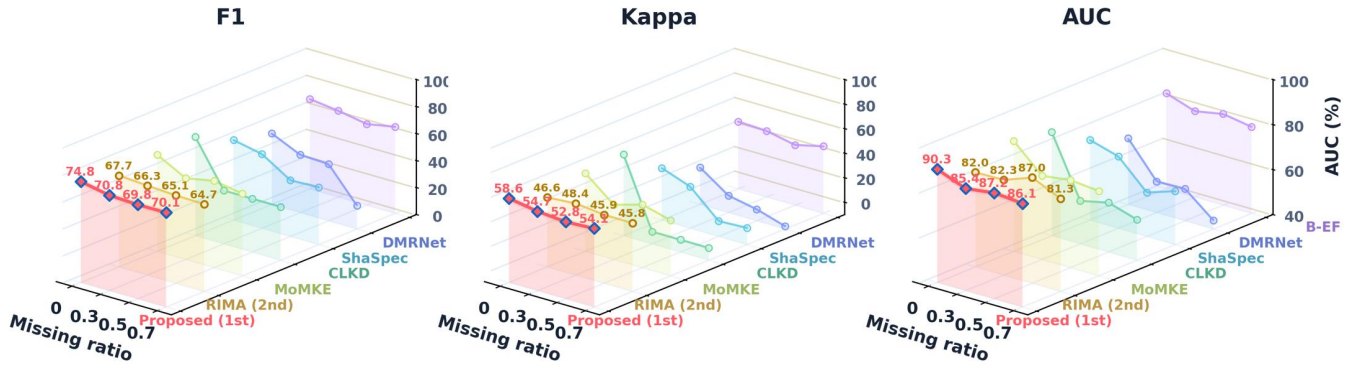


Fig. 4: Sensitivity to training-time missing ratios on three multimodal benchmarks. Performance trends of different methods under increasing proportions of missing modalities in the training set (0%, 30%, 50%, and 70%) on Harvard-30k AMD Luo et al. (2023), MM-ECG, and Derm7pt Kawahara et al. (2018), measured by F1-score, Cohen’s Kappa, and AUC. The proposed method shows stable performance across varying missing ratios, suggesting robustness to incomplete multimodal training data.

modality is randomly missing from each sample in both the training and test sets according to the specified missing ratio. As shown in Figure 4, most methods exhibit noticeable performance fluctuations as the missing ratio increases, whereas our method remains relatively stable across the three datasets. On Harvard-30k AMD, the proposed method maintains consistent performance under different missing ratios, particularly in terms of AUC, remaining above 82.7% across all settings and reaching 86.6% at 30% missingness. On MM-ECG, competitive methods are generally less sensitive to missingness, likely due to the strong correspondence between ECG signals and

images; nevertheless, our method still preserves stable performance (F1: 90.2-91.2%, Kappa: 84.7-86.4%, and AUC: 96.6-96.9%) across all ratios. On Derm7pt, which involves larger appearance variation and weaker cross-modal redundancy, the benefit of robust incomplete-modality modeling becomes more evident, and our method consistently remains among the top-performing approaches with only limited variation as the missing ratio increases. These results suggest that the proposed semantic decoupling and graph-based prototype completion strategy can effectively mitigate the adverse effect of training-time missing modalities and yield robust multimodal representations

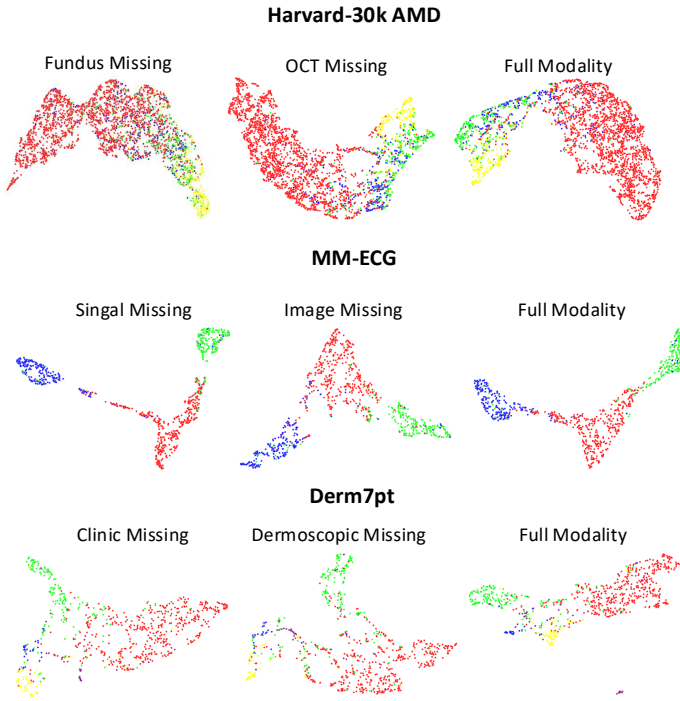


Fig. 5: t-SNE visualization of multimodal representations under different missing-modality conditions before the final Fully Connected layer.

(a) **Harvard-30k AMD** Luo et al. (2023): distributions under Fundus Missing, OCT Missing, and Full Modality settings. ● = Normal, ● = Early AMD, ● = intermediate AMD, ● = Late AMD, (b) **MM-ECG**: distributions under Signal Missing, Image Missing, and Full Modality settings. ● = Atrial Fibrillation, ● = Sinus Bradycardia, ● = Sinus Tachycardia, (c) **Derm7pt** Kawahara et al. (2018): distributions under Clinic Missing, Dermoscopic Missing, and Full Modality settings. ● = Nevus, ● = Basal Cell Carcinoma, ● = Melanoma, ● = Miscellaneous, ● = Seborrheic Keratosis.

even when incomplete samples are prevalent.

To further validate the classification effectiveness of the proposed framework, we visualize the learned multimodal representations under different missing-modality conditions using t-SNE, as shown in Figure 5. Across all three datasets (Harvard-30k AMD Luo et al. (2023), MM-ECG, and Derm7pt Kawahara et al. (2018)), our method yields discernible class tendencies consistent with clinical categories, even under missing-modality conditions. Although the class imbalance in Harvard-30k AMD Luo et al. (2023) and Derm7pt Kawahara et al. (2018) poses challenges to feature separation, our framework still preserves reasonable inter-class distinctions, particularly under missing-modality conditions.

#### 4.3. Visualization of Text-guided Semantic-Modality Decoupling

To evaluate the effectiveness of the proposed Text-guided Semantic-Modality Decoupling, we visualize the semantic feature  $s^m$  and the modality feature  $z^m$  from each dataset (Harvard-30k AMD Luo et al. (2023), MM-ECG, and Derm7pt Kawahara et al. (2018)). We additionally compare our model with two representative baselines: DMRNet Wei et al. (2024), a typical multi-modal learning framework, and RIMA Yu et al. (2025), a representative approach for incomplete-modality learning. We

use Grad-CAM Selvaraju et al. (2017) for all methods to ensure a consistent and fair visualization protocol. The visualizations for each dataset are shown from left to right in Figure 3. We observe that in cases where modality distinctions are pronounced (e.g., OCT and dermoscopic images), the semantic features consistently highlight pathology-related structures and lesion-relevant regions. And the modality features emphasize texture patterns, acquisition artifacts, or structural style, indicating a clear separation between diagnostic semantics and modality appearance. For fundus images, where semantic and stylistic cues are more entangled, our model still yields coherent semantic maps centered on disease-relevant retinal regions. Remarkably, on the ECG dataset, the semantic features concentrate on clinically meaningful waveform segments (e.g., QRS complexes and ST segments), whereas the modality features reflect amplitude and baseline variations.

In contrast, DMRNet Wei et al. (2024) is dominated by its multi-modal fusion pipeline. As a result, it often produces feature responses biased toward the modality with stronger visual cues, leading to modality-dependent and less reliable semantic attention. RIMA Yu et al. (2025), although designed for incomplete-modality learning, lacks an explicit semantic-modality disentanglement mechanism; consequently, its feature responses tend to mix semantic and stylistic cues, resulting in less accurate and less interpretable localization.

These observations collectively confirm that the proposed model achieves effective and interpretable semantic-modality decoupling across diverse modality types and imaging domains.

#### 4.4. Ablation Study of Proposed Components

To assess the effectiveness of our model, we conducted a thorough ablation study across three dataset, shown in Table 2. We evaluated the performance of the model by removing the Text-guided Semantic-Modality Decoupling, Labeled OT matching, GCN individually, and remaining the rest of the model structure unchanged. When removing labeled OT, we replace the class-consistent OT coupling with a naïve one-to-one matching. Similarly, when removing the GCN, we substitute it with a simple linear layer. Across all datasets, removing the semantic-modality decoupling leads to the most severe degradation. For example, in the Inter-Modality Missing setting on Harvard-30k AMD, the F1 score drops from 74.49 to 60.26 (OCT) and from 70.39 to 55.98 (Fundus). A similar trend is observed in MM-ECG, where removing decoupling reduces the F1 score from 93.01 to 91.35 (Signal) and from 89.49 to 85.93 (Image). The effect is even more pronounced on Derm7pt Kawahara et al. (2018), where the F1 score decreases from 73.45 to 53.11 (Clinical) and from 64.63 to 43.55 (Dermoscopy). These consistent drops indicate that semantic-modality disentanglement is essential for stable alignment under missing-modality conditions. Removing the labeled OT module leads to notable performance drops. For example, the F1 score decreases by 1.6% on Harvard-30k AMD, 0.7% in AUC on MM-ECG, and as much as 19.0% in Kappa on Derm7pt Kawahara et al. (2018). Ablating the GCN also degrades performance, with reductions of 0.8% in AUC on MM-ECG and 13.5% in Kappa on Derm7pt Kawahara et al. (2018).

Finally, when all three components are present, the model achieves the strongest performance across all datasets and in both missing-modality and complete-modality conditions. In

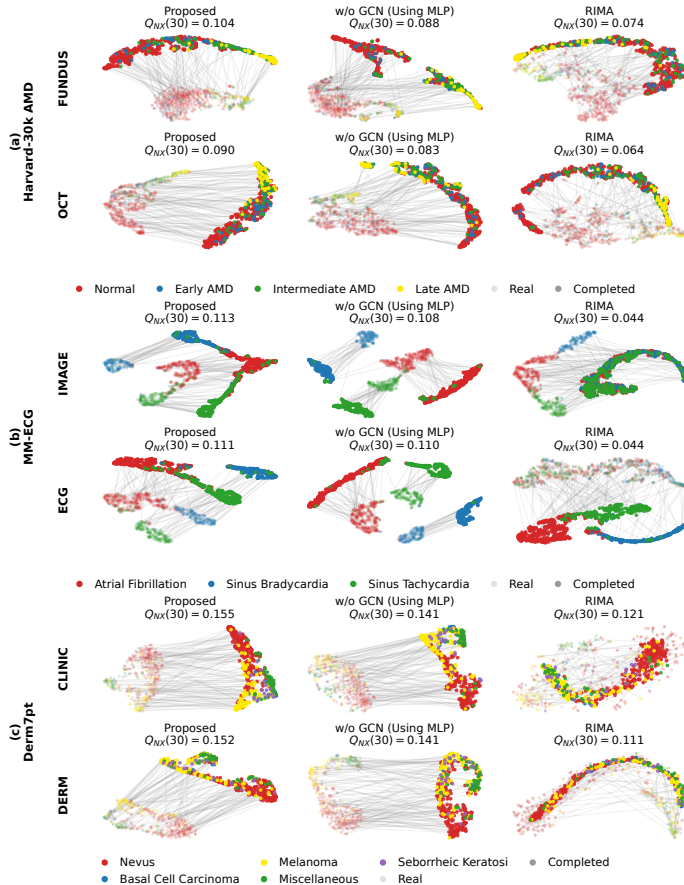


Fig. 6: t-SNE visualization of real and completed modality features with neighborhood preservation scores. We report  $Q_{NX}(30)$  to quantify top-30 local-neighborhood preservation between the real and completed feature spaces. (a) Harvard-30k AMD Luo et al. (2023), (b) MM-ECG, and (c) Derm7pt Kawahara et al. (2018). Rows denote modalities and columns compare the proposed model, w/o GCN (Using MLP), and RIMA. Colored points indicate class labels, translucent points are real features, solid points are completed features, and gray lines link paired samples. The proposed model consistently achieves higher  $Q_{NX}(30)$ , showing better preservation of modality-specific local structure after completion.

addition to the task-level ablation results, we further examine whether the completed modality features preserve the local geometry of the corresponding real modality feature space. As shown in Figure 6, we visualize the real and completed modality features using t-SNE, and further quantify their local-geometry consistency with  $Q_{NX}(30)$  Lee and Verleysen (2009), which measures the preservation of top-30 nearest-neighbor relationships between the two feature spaces. Across Harvard-30k AMD Luo et al. (2023), MM-ECG, and Derm7pt Kawahara et al. (2018), the proposed model consistently achieves higher neighborhood preservation than both RIMA and the variant without GCN. For example, on Harvard-30k AMD Luo et al. (2023), the proposed model improves  $Q_{NX}(30)$  from 0.088 to 0.104 on fundus and from 0.083 to 0.090 on OCT compared with the MLP-based variant. Similar gains are observed on Derm7pt Kawahara et al. (2018), where  $Q_{NX}(30)$  increases

from 0.141 to 0.155 for clinical features and from 0.141 to 0.152 for dermoscopic features. These results indicate that the proposed components, especially the graph-based semantic propagation, help preserve modality-specific neighborhood structure after completion, leading to more reliable representations under missing-modality conditions.

These experimental results demonstrate the effectiveness of each component based on different modality properties.

#### 4.5. Prompt robustness analysis.

Since the text embeddings are involved in both guide semantic-modality decoupling and adjacency matrix construction, we further investigate whether the performance is sensitive to the exact wording of class prompts. Specifically, we compare two prompt formulations: 1) an original modality-conditioned class prompt, which describes each category using a simple template such as “This is a clinic images. Nevus”; and 2) a detailed clinical prompt, which provides more explicit modality-specific diagnostic descriptions, such as “Clinical photograph of a melanocytic nevus presenting as a well-circumscribed pigmented macule”. For each dataset, the prompt pairs describe the same class labels and modality types, while differing only in the level of clinical detail. To isolate the effect of prompt wording, all other experimental settings, including data splits, model architecture, training schedule, and random seed, are kept unchanged. As shown in Table 3, the two prompt formulations lead to comparable overall performance across Harvard-30k AMD, MM-ECG, and Derm7pt under both complete-modality and inter-modality missing settings. Although the proposed framework generally maintains comparable performance under different prompt formulations, minor-to-moderate variations are observed in certain missing-modality settings. This suggests that prompt wording may affect the estimated semantic relations to some extent, while the overall framework remains robust across prompt designs. Future work may explore prompt ensembling or learnable prompt optimization to further improve stability.

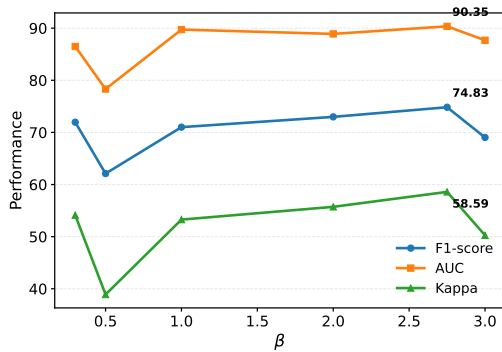
#### 4.6. Loss Weight Sensitivity Analysis

To investigate the influence of loss weight configurations on model performance, we conduct a sensitivity analysis on the weighting coefficients of the disentanglement loss ( $\beta$ ) and the completion loss ( $\gamma$ ), while fixing the cross-entropy loss weight to 1. Specifically,  $\beta$  controls the contribution of the semantic-modality disentanglement objective  $\mathcal{L}_{\text{decouple}}$ , and  $\gamma$  balances the prototype-based completion objective  $\mathcal{L}_{\text{com}}$  in the overall loss function (Equation 15).

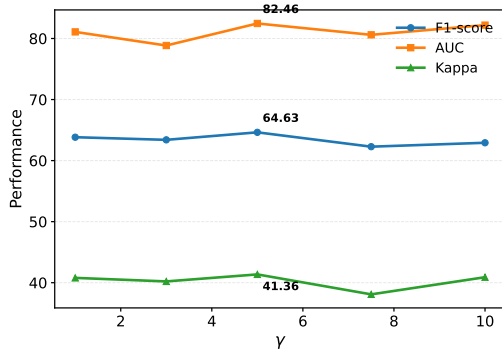
**Effect of  $\beta$ .** We first conduct the sensitivity analysis of  $\beta$  on the Derm7pt Kawahara et al. (2018) dataset under the *Complete-Modality Fusion* setting, where all modalities are available. In this case, the completion loss  $\mathcal{L}_{\text{com}}$  does not affect classification results, allowing us to fix  $\gamma$  and isolate the effect of the disentanglement loss weight. As shown in Figure 7a, model performance improves as  $\beta$  increases, and reaches its optimum around  $\beta = 2.75$ , achieving an F1-score of 69.34 %, Kappa of 53.18 %, and AUC of 88.42 %.

Table 3: Prompt robustness analysis across three datasets, including Harvard-30k AMD Luo *et al.* (2023), MM-ECG, and Derm7pt Kawahara *et al.* (2018). We compare the original modality-conditioned class prompts with more detailed clinical prompts.

| Prompt Setting                  | Harvard-30k AMD Luo <i>et al.</i> (2023) |        |       |       |       | MM-ECG |       |       |       |       | Derm7pt Kawahara <i>et al.</i> (2018) |        |       |       |       |
|---------------------------------|------------------------------------------|--------|-------|-------|-------|--------|-------|-------|-------|-------|---------------------------------------|--------|-------|-------|-------|
|                                 | OCT                                      | Fundus | Kappa | AUC   | F1    | Signal | Image | Kappa | AUC   | F1    | Clinic                                | Dermo. | Kappa | AUC   | F1    |
| <b>Inter-Modality Missing</b>   |                                          |        |       |       |       |        |       |       |       |       |                                       |        |       |       |       |
| Original class prompt           |                                          | ✓      | 46.44 | 84.13 | 74.49 |        | ✓     | 88.85 | 97.58 | 93.01 |                                       | ✓      | 57.71 | 87.90 | 73.45 |
| Detailed clinical prompt        |                                          | ✓      | 56.02 | 90.11 | 69.67 |        | ✓     | 89.74 | 96.21 | 93.58 |                                       | ✓      | 59.91 | 88.13 | 74.69 |
| Original class prompt           | ✓                                        |        | 44.57 | 80.99 | 70.39 | ✓      |       | 83.56 | 96.32 | 89.49 | ✓                                     |        | 41.36 | 82.46 | 64.63 |
| Detailed clinical prompt        | ✓                                        |        | 33.48 | 72.32 | 66.52 | ✓      |       | 90.46 | 97.14 | 94.01 | ✓                                     |        | 31.39 | 80.08 | 58.42 |
| <b>Complete-Modality Fusion</b> |                                          |        |       |       |       |        |       |       |       |       |                                       |        |       |       |       |
| Original class prompt           | ✓                                        | ✓      | 50.92 | 84.84 | 75.28 | ✓      | ✓     | 88.10 | 97.58 | 92.74 | ✓                                     | ✓      | 58.59 | 90.35 | 74.83 |
| Detailed clinical prompt        | ✓                                        | ✓      | 49.86 | 79.47 | 75.13 | ✓      | ✓     | 90.06 | 97.66 | 94.00 | ✓                                     | ✓      | 52.88 | 89.37 | 71.52 |



(a) Effect of  $\beta$  (Derm7pt Kawahara *et al.* (2018), complete-modality fusion)



(b) Effect of  $\gamma$  (Derm7pt Kawahara *et al.* (2018), inter-modality missing)

Fig. 7: Sensitivity analysis of loss weighting coefficients on the Derm7pt Kawahara *et al.* (2018) dataset.

**Effect of  $\gamma$ .** Based on the optimal  $\beta$  value, we then evaluate the effect of the completion loss weight  $\gamma$  under the *Inter-Modality Missing* setting. We test the case where a clinic image is missing. Since the magnitude of  $\mathcal{L}_{com}$  is smaller compared to  $\mathcal{L}_{ce}$  and  $\mathcal{L}_{decouple}$ , we explore  $\gamma$  values ranging from 1 to 10. As illustrated in Figure 7b, model performance improves steadily and reaches the best trade-off at  $\gamma = 5$ , with an F1-score of 64.63 %, Kappa of 41.36 %, and AUC of 82.46 %.

## 5. Discussion

Although the loss weighting coefficients were selected via grid search and consistently achieved optimal performance across all three evaluated datasets, we do not exclude the possibility that different datasets, modality combinations, or downstream tasks may require re-tuning of these parameters. In addition, the current study primarily focuses on single-modality missing scenarios, which are common in clinical practice, while more complex missing scenarios remain to be explored. Moreover, although the learned semantic representations exhibit strong consistency with clinically meaningful patterns in the evaluated datasets, it remains to be demonstrated that these semantic features are universally equivalent to expert clinical reasoning across untested diseases, imaging protocols, and clinical contexts.

These limitations also point to several directions for future work, including extending the proposed framework beyond classification to clinically prioritized downstream tasks such as prognosis and survival risk prediction; adapting the semantic-modality decoupling and prototype completion mechanisms to dense prediction tasks such as medical image segmentation; incorporating richer language supervision or knowledge-enhanced representations to further strengthen semantic alignment, interpretability, and generalization across diverse clinical scenarios.

## 6. Conclusion

We proposed a novel multimodal alignment and fusion framework designed to handle both complete and incomplete modality scenarios in medical imaging. By decoupling diagnosis-relevant semantics from modality-specific variations, we establish class-consistent cross-modal alignment via labeled optimal transport. Furthermore, the text-level OT induced relational graph enables graph-based prototype completion, allowing the model to semantically reconstruct missing modality features without relying on explicit image synthesis. Extensive experiments across three heterogeneous multimodal datasets demonstrate that our framework consistently outperforms state-of-the-art methods in both modality-fusion and missing-modality settings.

## Author Contributions

Qinkai Yu: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Visualization, Writing – original draft. Jing Li: Methodology, Software, Validation, Writing – review & editing. Feixiang Zhou: Validation, Writing – review & editing. Jianyang Xie: Validation, Writing – review & editing. Yitian Zhao: Resources, Writing – review & editing. Yalin Zheng: Supervision, Resources, Writing – review & editing. Aline Villavicencio: Supervision, Writing – review & editing. Gregory Y.H. Lip: Supervision, Clinical interpretation, Writing – review & editing. Yanda Meng: Conceptualization, Supervision, Project administration, Funding acquisition, Writing – review & editing.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Code availability

The implementation code will be made publicly available upon acceptance.

## References

- Chen, G., Yao, W., Song, X., Li, X., Rao, Y., Zhang, K., 2022. Plot: Prompt learning with optimal transport for vision-language models. *arXiv preprint arXiv:2210.01253*.
- Chen, Q., Zhang, J., Meng, R., Zhou, L., Li, Z., Feng, Q., Shen, D., 2024. Modality-specific information disentanglement from multi-parametric mri for breast tumor segmentation and computer-aided diagnosis. *IEEE Transactions on Medical Imaging* 43, 1958–1971.
- Cuturi, M., 2013. Sinkhorn distances: Lightspeed computation of optimal transport. *Advances in neural information processing systems* 26.
- Du, C., Teng, J., Li, T., Liu, Y., Yuan, T., Wang, Y., Yuan, Y., Zhao, H., 2023. On uni-modal feature learning in supervised multi-modal learning, in: *International Conference on Machine Learning*, PMLR. pp. 8632–8656.
- Du, S., Luo, X., O'Regan, D.P., Qin, C., 2025. STiL: Semi-supervised tabular-image learning for comprehensive task-relevant information exploration in multimodal classification, in: *Conference on Computer Vision and Pattern Recognition (CVPR)* 2025.
- Gazdieva, M., Korotin, A., Selikhanovych, D., Burnaev, E., 2023. Extremal domain translation with neural optimal transport. *Advances in Neural Information Processing Systems* 36, 40381–40413.
- Haribabu, M., Guruviah, V., Yogarajah, P., 2023. Recent advancements in multimodal medical image fusion techniques for better diagnosis: an overview. *Current Medical Imaging Reviews* 19, 673–694.
- He, K., Zhang, X., Ren, S., Sun, J., 2016. Deep residual learning for image recognition, in: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pp. 770–778.
- Hua, C.H., Kim, K., Huynh-The, T., You, J.I., Yu, S.Y., Le-Tien, T., Bae, S.H., Lee, S., 2020. Convolutional network with twofold feature augmentation for diabetic retinopathy recognition from multi-modal images. *IEEE Journal of Biomedical and Health Informatics* 25, 2686–2697.
- Jin, J., Wang, H., Li, H., Li, J., Pan, J., Hong, S., 2025. Reading your heart: Learning ecg words and sentences via pre-training ecg language model. *arXiv preprint arXiv:2502.10707*.
- Johnson, A.E., Bulgarelli, L., Shen, L., Gayles, A., Shammout, A., Horng, S., Pollard, T.J., Hao, S., Moody, B., Gow, B., et al., 2023. MIMIC-IV, a freely accessible electronic health record dataset. *Scientific data* 10, 1.

- Kawahara, J., Daneshvar, S., Argenziano, G., Hamarneh, G., 2018. Seven-point checklist and skin lesion classification using multitask multimodal neural nets. *IEEE journal of biomedical and health informatics* 23, 538–546.
- Lee, J.A., Verleysen, M., 2009. Quality assessment of dimensionality reduction: Rank-based criteria. *Neurocomputing* 72, 1431–1443.
- Li, Y., Daho, M.E.H., Conze, P.H., Zeghlache, R., Le Boité, H., Tadayoni, R., Cochener, B., Lamard, M., Quéllec, G., 2024a. A review of deep learning-based information fusion techniques for multimodal medical image classification. *Computers in Biology and Medicine* 177, 108635.
- Li, Z., Zhang, Y., Li, H., Chai, Y., Yang, Y., 2024b. Deformation-aware and reconstruction-driven multimodal representation learning for brain tumor segmentation with missing modalities. *Biomedical Signal Processing and Control* 91, 106012.
- Liu, C., Huang, Z., Chen, Z., Tang, F., Tian, Y., Xu, Z., Luo, Z., Zheng, Y., Meng, Y., 2025. Incomplete modality disentangled representation for ophthalmic disease grading and diagnosis, in: *Proceedings of the AAAI Conference on Artificial Intelligence*, pp. 5361–5369.
- Liu, R., Bai, Y., Yue, X., Zhang, P., 2024. Teach multimodal llms to comprehend electrocardiographic images. *arXiv preprint arXiv:2410.19008*.
- Loshchilov, I., Hutter, F., 2017. Decoupled weight decay regularization. *arXiv preprint arXiv:1711.05101*.
- Luo, Y., Tian, Y., Shi, M., Elze, T., Wang, M., 2023. Eye fairness: A large-scale 3d imaging dataset for equitable eye diseases screening and fair identity scaling. *arXiv preprint arXiv:2310.02492*.
- Luong, M., Nguyen, K., Ho, N., Haf, R., Phung, D., Qu, L., 2024. Revisiting deep audio-text retrieval through the lens of transportation. *arXiv preprint arXiv:2405.10084*.
- Meng, Y., Zhang, Y., Xie, J., Duan, J., Joddrell, M., Madhusudhan, S., Peto, T., Zhao, Y., Zheng, Y., 2024. Multi-granularity learning of explicit geometric constraint and contrast for label-efficient medical image segmentation and differentiable clinical function assessment. *Medical Image Analysis*.
- Park, J., Lee, J., Sohn, K., 2024. Bridging vision and language spaces with assignment prediction. *arXiv preprint arXiv:2404.09632*.
- Peyré, G., Cuturi, M., Solomon, J., 2016. Gromov-wasserstein averaging of kernel and distance matrices, in: *International conference on machine learning*, PMLR. pp. 2664–2672.
- Peyré, G., Cuturi, M., et al., 2019. Computational optimal transport: With applications to data science. *Foundations and Trends® in Machine Learning* 11, 355–607.
- Pipoli, V., Saporita, A., Marchesini, K., Grana, C., Ficarra, E., Bolelli, F., 2025. Im-fuse: A mamba-based fusion block for brain tumor segmentation with incomplete modalities, in: *International Conference on Medical Image Computing and Computer-Assisted Intervention*, Springer. pp. 225–235.
- Radford, A., Kim, J.W., Hallacy, C., Ramesh, A., Goh, G., Agarwal, S., Sastry, G., Askell, A., Mishkin, P., Clark, J., et al., 2021. Learning transferable visual models from natural language supervision, in: *International conference on machine learning*, PMLR. pp. 8748–8763.
- Ryu, J., Lopez, R., Bunne, C., Regev, A., 2024. Cross-modality matching and prediction of perturbation responses with labeled gromov-wasserstein optimal transport. *arXiv:2405.00838*.
- Selvaraju, R.R., Cogswell, M., Das, A., Vedantam, R., Parikh, D., Batra, D., 2017. Grad-cam: Visual explanations from deep networks via gradient-based localization, in: *Proceedings of the IEEE international conference on computer vision*, pp. 618–626.
- Shivashankara, K.K., Shervedani, A.M., Clifford, G.D., Reyna, M.A., Sameni, R., et al., 2024. Ecg-image-kit: a synthetic image generation toolbox to facilitate deep learning-based electrocardiogram digitization. *Physiological measurement* 45, 055019.
- Sinkhorn, R., Knopp, P., 1967. Concerning nonnegative matrices and doubly stochastic matrices. *Pacific Journal of Mathematics* 21, 343–348.
- Tang, W., He, F., Liu, Y., Duan, Y., 2022. Matr: Multimodal medical image fusion via multiscale adaptive transformer. *IEEE Transactions on Image Processing* 31, 5134–5149.
- Tran, D., Wang, H., Torresani, L., Ray, J., LeCun, Y., Paluri, M., 2018. A closer look at spatiotemporal convolutions for action recognition, in: *Proceedings of the IEEE conference on Computer Vision and Pattern Recognition*, pp. 6450–6459.
- Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A.N., Kaiser, L., Polosukhin, I., 2017. Attention is all you need, in: *NIPS*.
- Wang, H., Chen, Y., Ma, C., Avery, J., Hull, L., Carneiro, G., 2023a. Multimodal learning with missing modality via shared-specific feature modelling, in: *Proceedings of the IEEE/CVF conference on computer vision and pattern*

- recognition, pp. 15878–15887.
- Wang, L., Dai, W., Jin, M., Ou, C., Li, X., 2023b. Fundus-enhanced disease-aware distillation model for retinal disease classification from oct images, in: International Conference on Medical Image Computing and Computer-Assisted Intervention, Springer. pp. 639–648.
- Wang, L., Wang, M., Fu, H., Zhang, D., 2024. Say no to freeloader: Protecting intellectual property of your deep model. *IEEE Transactions on Pattern Analysis and Machine Intelligence* 46, 11073–11086.
- Wang, L., Wang, M., Zhang, D., Fu, H., 2023c. Model barrier: A compact untransferable isolation domain for model intellectual property protection, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 20475–20484.
- Watanabe, T., Hiratsuka, Y., Kita, Y., Tamura, H., Kawasaki, R., Yokoyama, T., Kawashima, M., Nakano, T., Yamada, M., 2022. Combining optical coherence tomography and fundus photography to improve glaucoma screening. *Diagnostics* 12, 1100.
- Wei, S., Luo, Y., Wang, Y., Luo, C., 2024. Robust multimodal learning via representation decoupling, in: European Conference on Computer Vision, Springer. pp. 38–54.
- Wohlrapp, M., Bubeck, N., Rueckert, D., Lotter, W., 2026. Evaluating the impact of medical image reconstruction on downstream ai fairness and performance, in: Medical Imaging with Deep Learning-Validation Papers.
- Wu, J., Fang, H., Li, F., Fu, H., Lin, F., Li, J., Huang, Y., Yu, Q., Song, S., Xu, X., et al., 2023. Gamma challenge: glaucoma grading from multi-modality images. *Medical Image Analysis* 90, 102938.
- Wu, R., Wang, H., Chen, H.T., Carneiro, G., 2024. Deep multimodal learning with missing modality: A survey. *arXiv preprint arXiv:2409.07825*.
- Xia, P., Hu, M., Tang, F., Li, W., Zheng, W., Ju, L., Duan, P., Yao, H., Ge, Z., 2024. Generalizing to unseen domains in diabetic retinopathy with disentangled representations, in: International Conference on Medical Image Computing and Computer-Assisted Intervention, Springer. pp. 427–437.
- Xu, F., Fu, P., Huang, Q., Zou, B., Aw, A., Wang, M., 2023. Leveraging contrastive learning and knowledge distillation for incomplete modality rumor detection, in: Findings of the Association for Computational Linguistics: EMNLP 2023, pp. 13492–13503.
- Xu, W., Jiang, H., Liang, X., 2024. Leveraging knowledge of modality experts for incomplete multimodal learning, in: Proceedings of the 32nd ACM International Conference on Multimedia, pp. 438–446.
- Xu, Y., Chen, H., 2023. Multimodal optimal transport-based co-attention transformer with global structure consistency for survival prediction, in: Proceedings of the IEEE/CVF international conference on computer vision, pp. 21241–21251.
- Yang, Y., Chen, H., Chang, Z., Xiang, Y., Ye, C., Ma, T., 2024. Incomplete learning of multi-modal connectome for brain disorder diagnosis via modal-mixup and deep supervision, in: Medical Imaging With Deep Learning, PMLR. pp. 1006–1018.
- Yao, W., Yin, K., Cheung, W.K., Liu, J., Qin, J., 2024. Drfuse: Learning disentangled representation for clinical multi-modal fusion with missing modality and modal inconsistency, in: Proceedings of the AAAI conference on artificial intelligence, pp. 16416–16424.
- Ye, W., Guo, Z., Ren, Y., Tian, Y., Shen, Y., Chen, Z., He, J., Ke, J., Shen, Y., 2025. Diffm 4 ri: a latent diffusion model with modality inpainting for synthesizing missing modalities in mri analysis. *IEEE Journal of Biomedical and Health Informatics*.
- Yu, Q., Xie, J., Zhao, Y., Chen, C., Zhang, L., Chen, L., Cheng, J., Liu, L., Zheng, Y., Meng, Y., 2025. Robust incomplete-modality alignment for ophthalmic disease grading and diagnosis via labeled optimal transport, in: International Conference on Medical Image Computing and Computer-Assisted Intervention, Springer. pp. 553–563.
- Zhao, F., Zhang, C., Geng, B., 2024. Deep multimodal data fusion. *ACM computing surveys* 56, 1–36.
- Zuo, Q., Zhang, J., Yang, Y., 2021. Dmc-fusion: Deep multi-cascade fusion with classifier-based feature synthesis for medical multi-modal images. *IEEE Journal of Biomedical and Health Informatics* 25, 3438–3449.